

Practice describing scatterplots

MATH 213

Instructions

In this activity, we will practice describing scatterplots.

In a moment, you will find your group's slides. Each group has 4 slides.

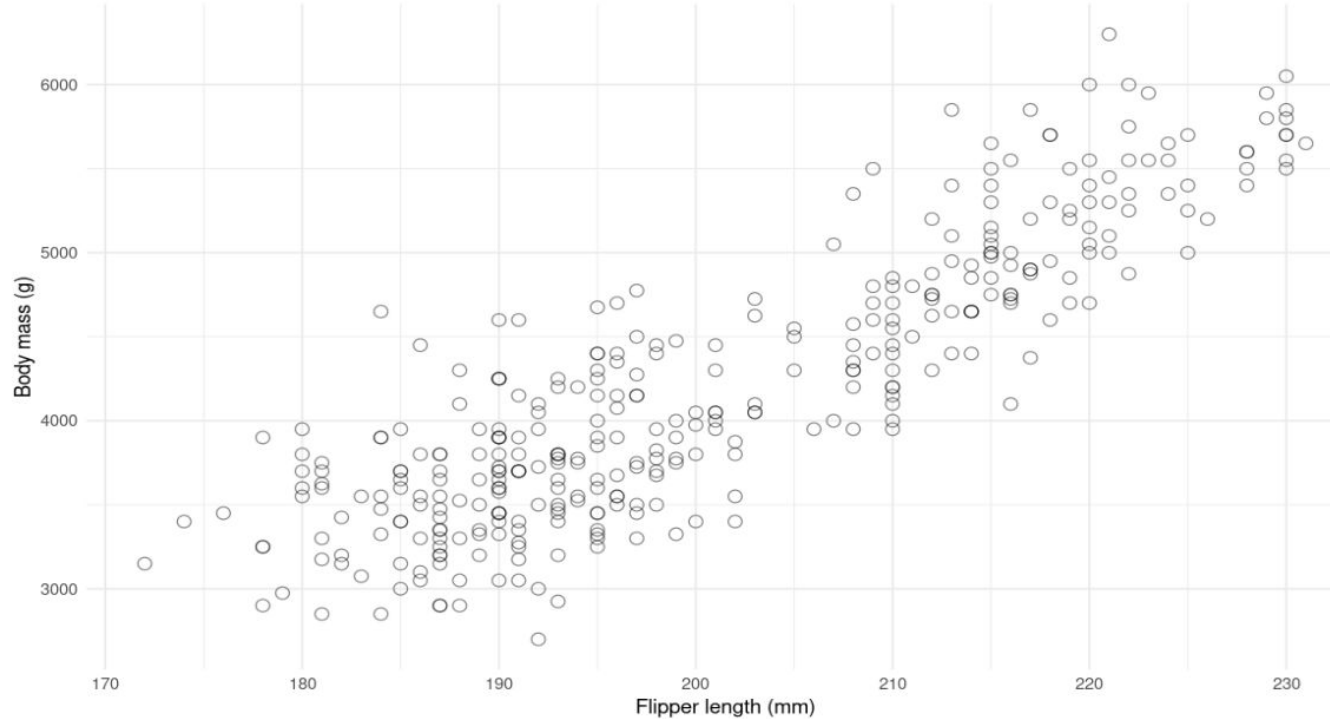
Each student should begin by taking a few minutes to think or write quietly in their own notebook to try to describe each scatterplot. (Set a timer for 2-3 minutes.)

Then, collaborate to write a nice description of each scatterplot on your group's slides.

Group 1 - Question 1

Researchers collected data on size measurements for three penguin species observed on three islands in the Palmer Archipelago, Antarctica. The data were collected from 2007-2009 as part of the US Long Term Ecological Research Network. Here is a graph of the penguins' body mass (in grams) versus their flipper length (in millimeters).

Penguin size, Palmer Station LTER
Flipper length and body mass for Adelie, Chinstrap and Gentoo Penguins

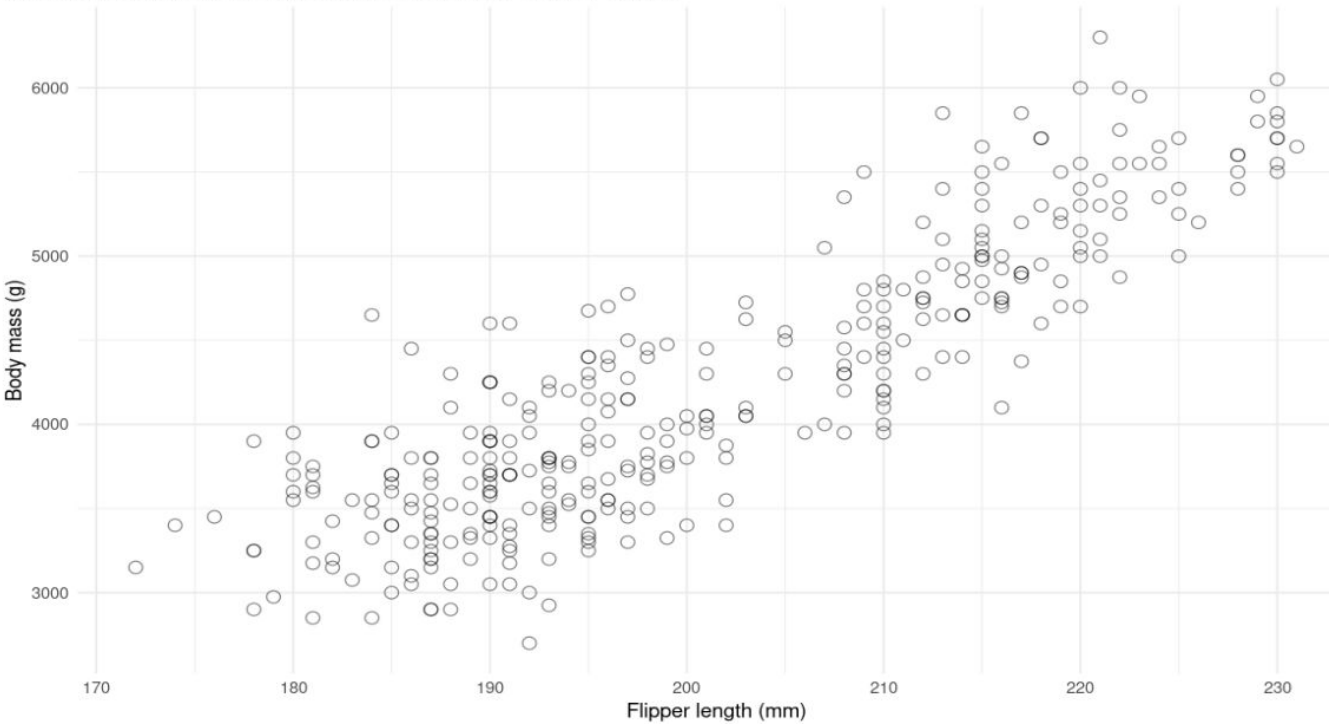


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Describe the relationship between the penguins' body mass and their flipper length as displayed in the scatterplot. Make sure to refer to the direction, strength and shape of the association, if any, and to speak in context.

The scatterplot shows a relationship between Penguin flipper length to Body mass. There is a strong increase trend in the graph. As the body mass of a penguin increases, the flipper length also increases. Looking towards the beginning, A penguin at a body mass of 3500g has a flipper length of 180. Now at 5,000, the flipper length reaches 215. These two points show that the strength of the graph is moderately consistent and the direction is positive.

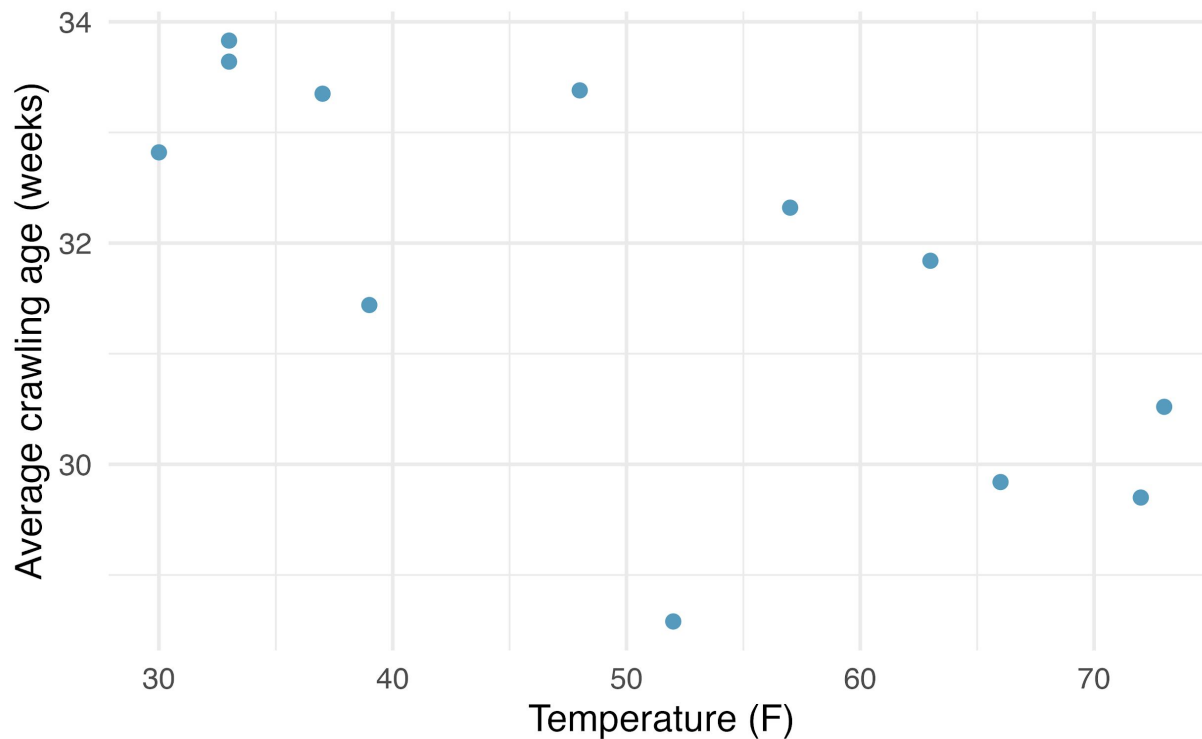
Penguin size, Palmer Station LTER
Flipper length and body mass for Adelie, Chinstrap and Gentoo Penguins



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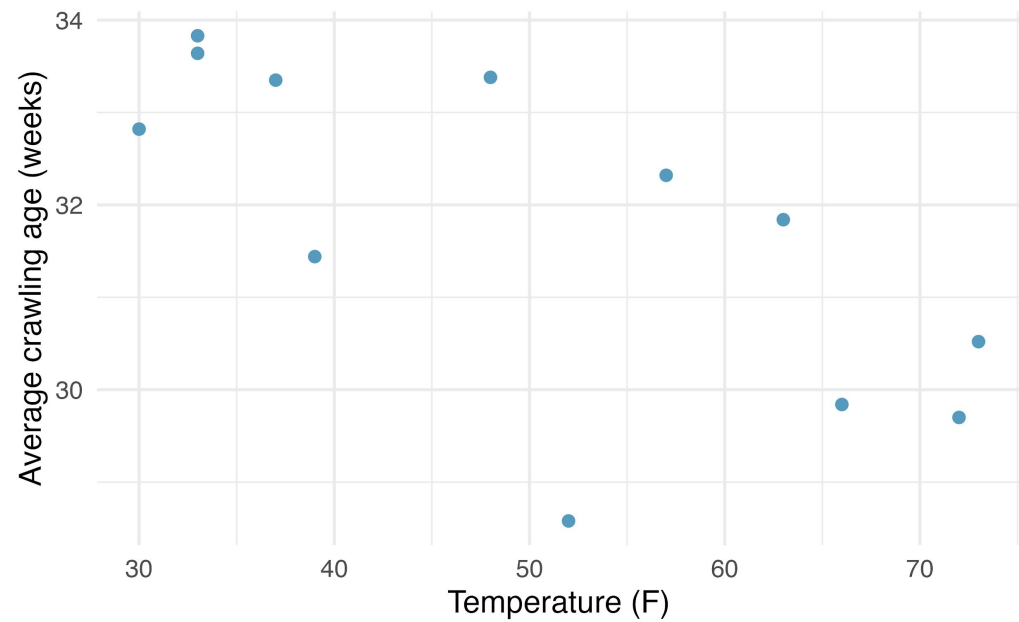
Question 2

Crawling babies. A study conducted at the University of Denver investigated whether babies take longer to learn to crawl in cold months, when they are often bundled in clothes that restrict their movement, than in warmer months. Infants born during the study year were split into twelve groups, one for each birth month. We consider the average crawling age of babies in each group against the average temperature when the babies are six months old (that's when babies often begin trying to crawl). Temperature is measured in degrees Fahrenheit (F) and age is measured in weeks. (Benson 1993)



Describe the relationship between average crawling age and average temperature as displayed in the scatterplot. Make sure to refer to the direction, strength and shape of the association, if any, and to speak in context.

There is a very weak connection between temperature and average crawling age. The linear connection between the two data points is slightly negative. Some points on the graph that show how weak the graph is are the points at 37,31.5 and 63, 31.7. These points demonstrate how weak the graph's correlation is by demonstrating a large temperature difference while still having a very similar average crawling age.

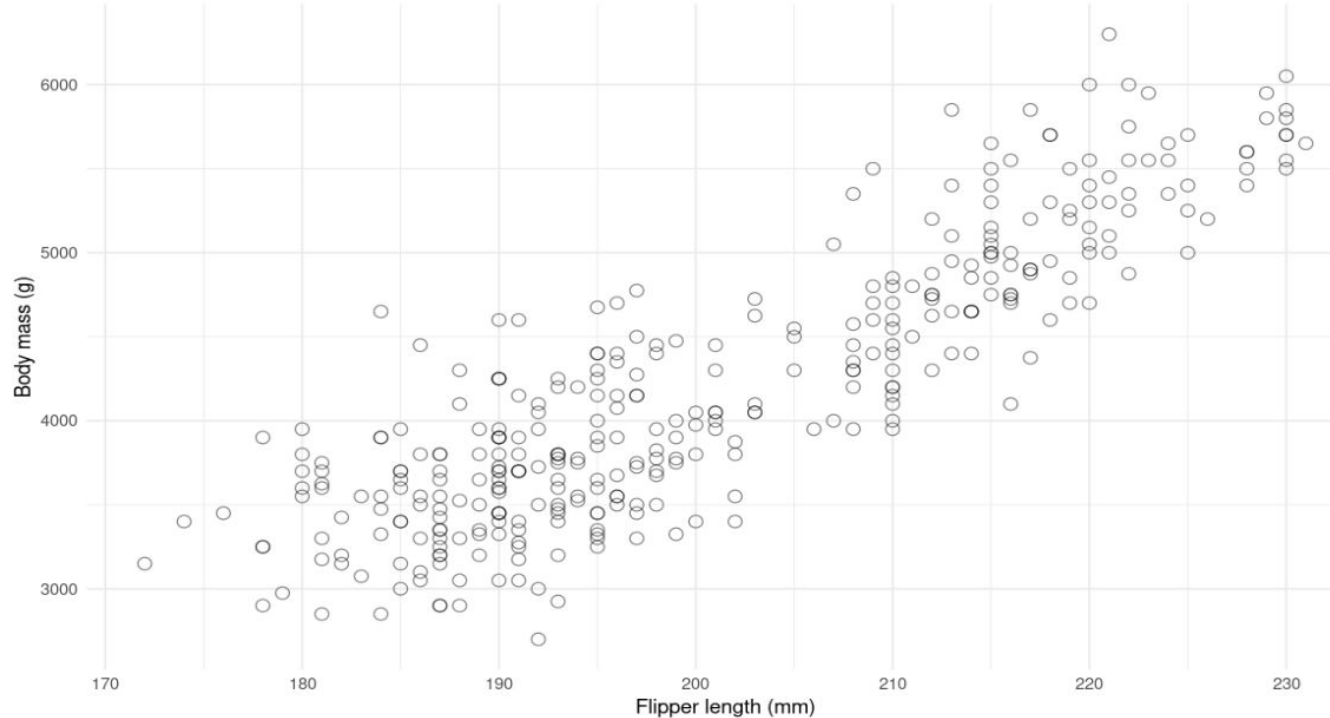


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Group 2 - Question 1

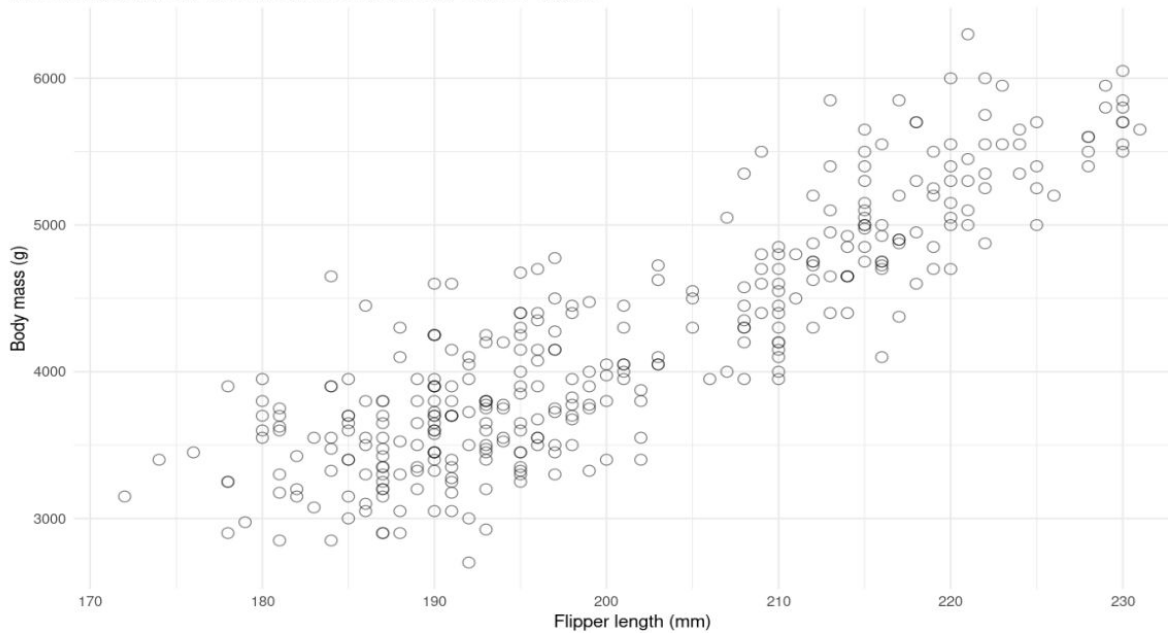
Researchers collected data on size measurements for three penguin species observed on three islands in the Palmer Archipelago, Antarctica. The data were collected from 2007-2009 as part of the US Long Term Ecological Research Network. Here is a graph of the penguins' body mass (in grams) versus their flipper length (in millimeters).

Penguin size, Palmer Station LTER
Flipper length and body mass for Adelie, Chinstrap and Gentoo Penguins



Go to the next slide to write your description.

Penguin size, Palmer Station LTER
Flipper length and body mass for Adelie, Chinstrap and Gentoo Penguins



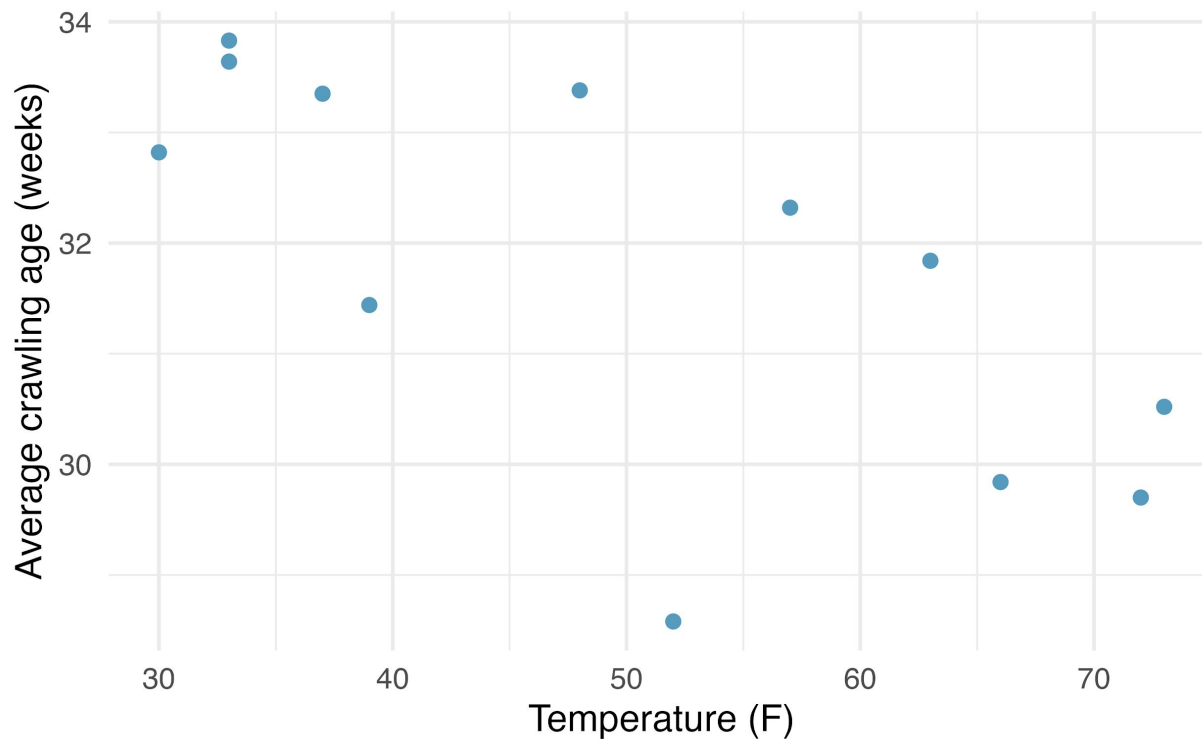
Describe the relationship between the penguins' body mass and their flipper length as displayed in the scatterplot. Make sure to refer to the direction, strength and shape of the association, if any, and to speak in context.

This scatterplot shows the relationship between the flipper length (mm) and body mass (g) of three different penguin species. Generally, the graph demonstrates a strong, linear, positive association meaning the longer the flipper length the larger the body mass of the penguin; those with shorter flipper lengths tend to have smaller body masses.

For example, a penguin with a flipper length of 220mm has a body mass of 6000 grams, and there is another penguin with a flipper length of 180mm that has a body mass of about 3000 grams. The relationship is strong, meaning there is little to no variation in flipper length for the penguins body mass. Penguins that have a flipper length of 190 mm have a body mass that ranges from about 3000 to 4000 grams.

Question 2

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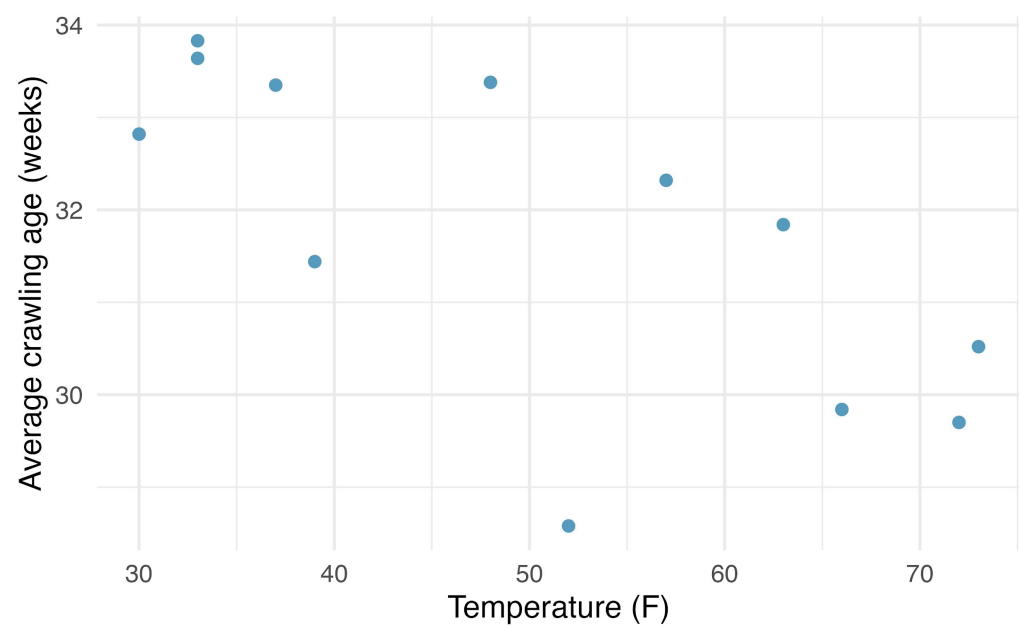


Describe the relationship between average crawling age and average temperature as displayed in the scatterplot. Make sure to refer to the direction, strength and shape of the association, if any, and to speak in context.

This scatterplot shows the relationship between the average crawling age in weeks compared to the average temperature.

There is a weak negative linear association between these variables: colder months cause babies to learn to crawl slower, and warmer months cause babies to learn to crawl faster.

For example, one group of babies took an average of about 33 weeks to learn how to crawl when it was about 30 degrees outside while another group took about 29 weeks when it waas 70 degrees outside.

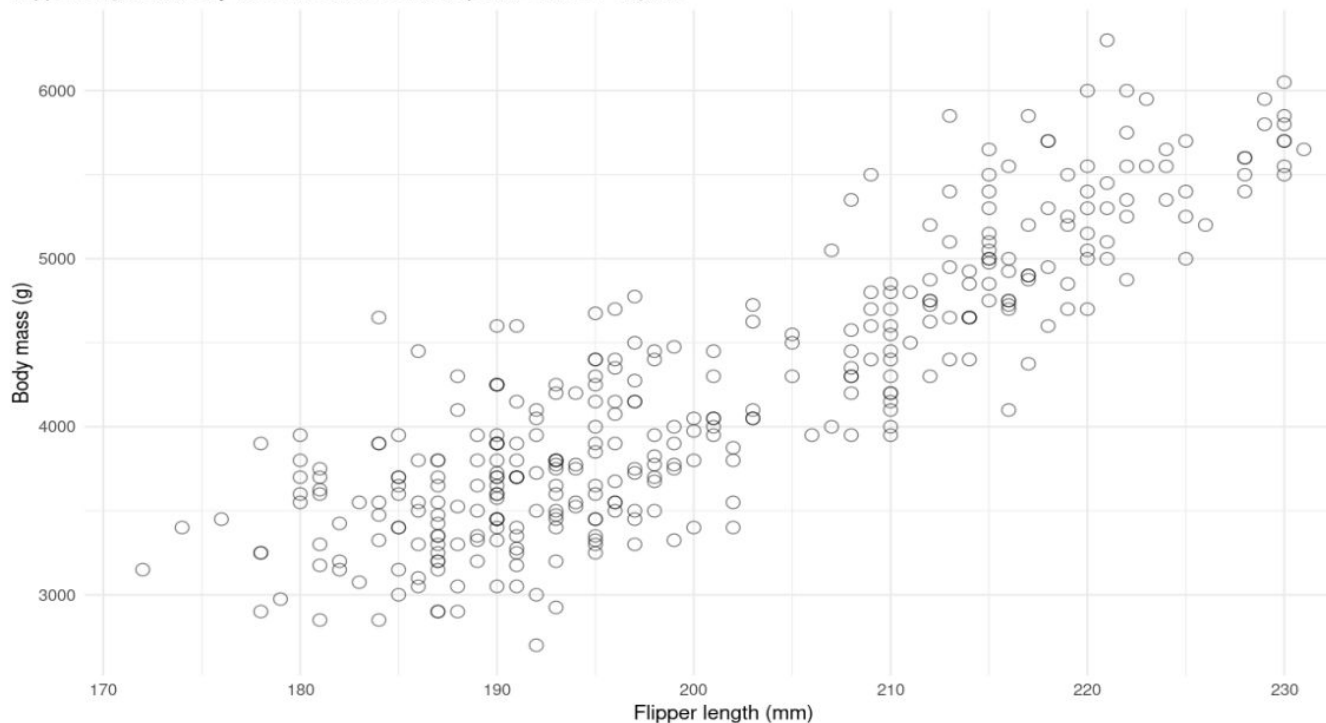


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Group 3 - Question 1

Researchers collected data on size measurements for three penguin species observed on three islands in the Palmer Archipelago, Antarctica. The data were collected from 2007-2009 as part of the US Long Term Ecological Research Network. Here is a graph of the penguins' body mass (in grams) versus their flipper length (in millimeters).

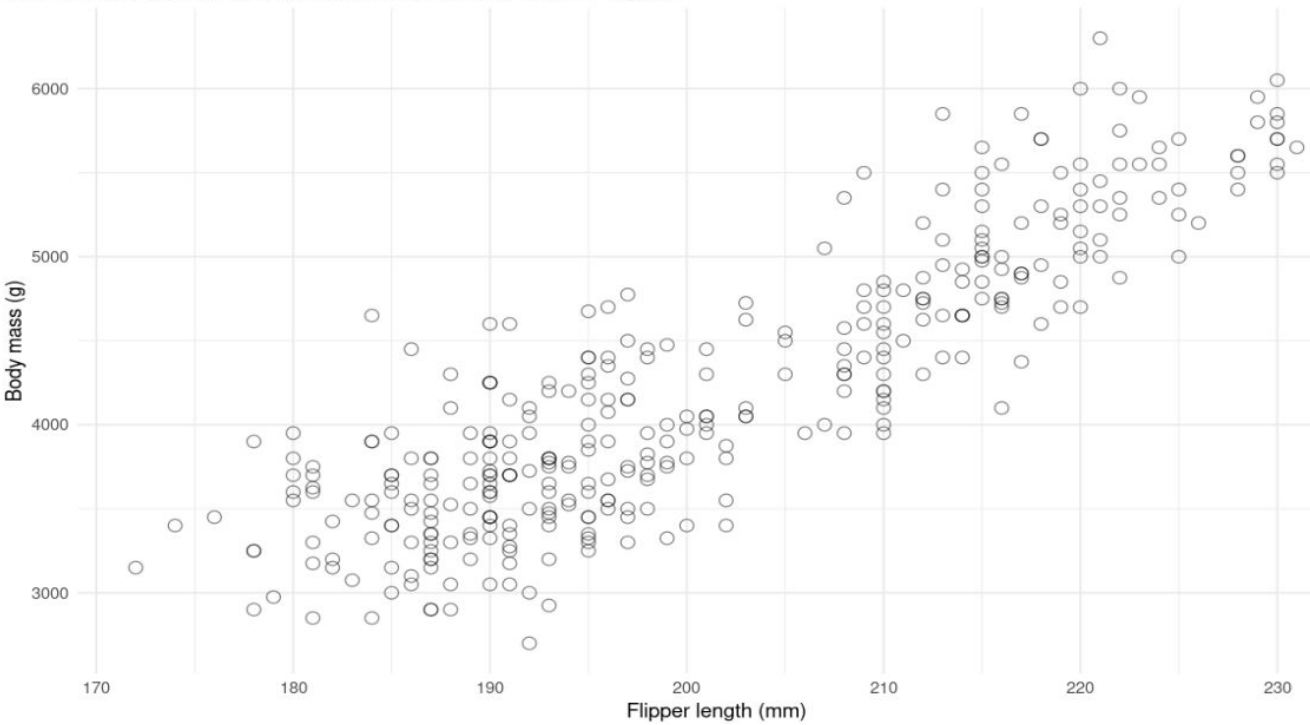
Penguin size, Palmer Station LTER
Flipper length and body mass for Adelie, Chinstrap and Gentoo Penguins



Go to the next slide to write your description.

Describe the relationship between the penguins' body mass and their flipper length as displayed in the scatterplot. Make sure to refer to the direction, strength and shape of the association, if any, and to speak in context.

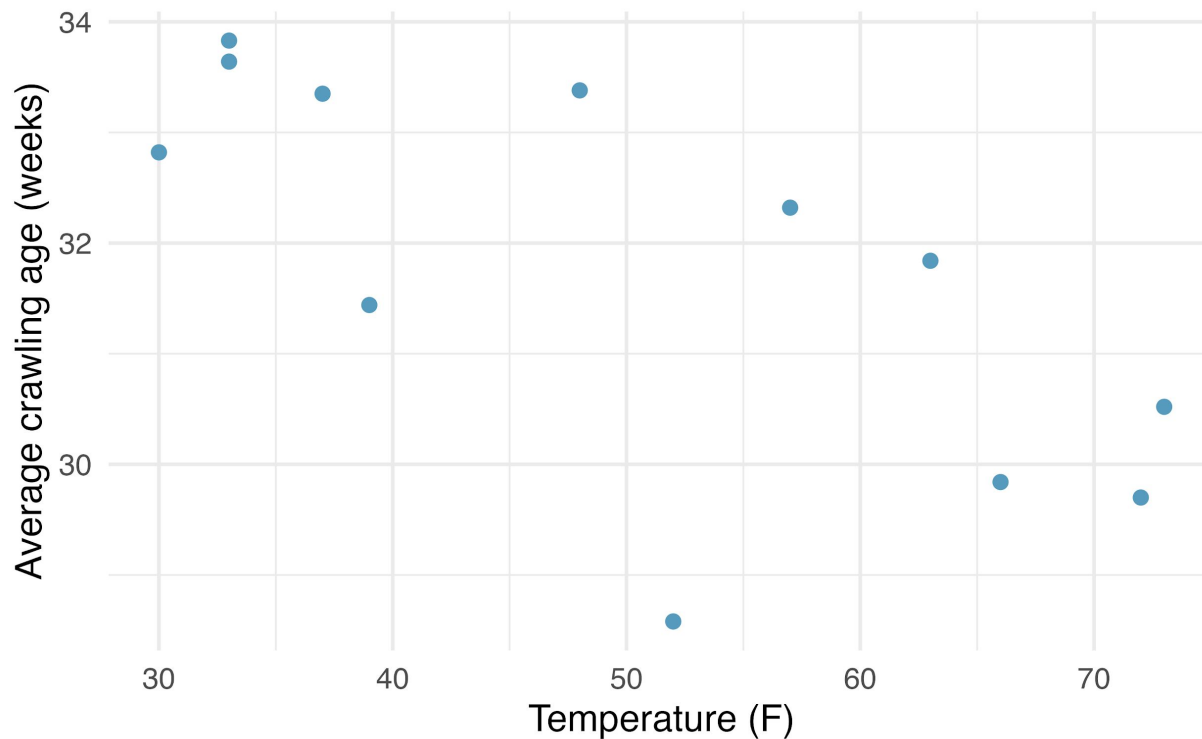
Penguin size, Palmer Station LTER
Flipper length and body mass for Adelie, Chinstrap and Gentoo Penguins



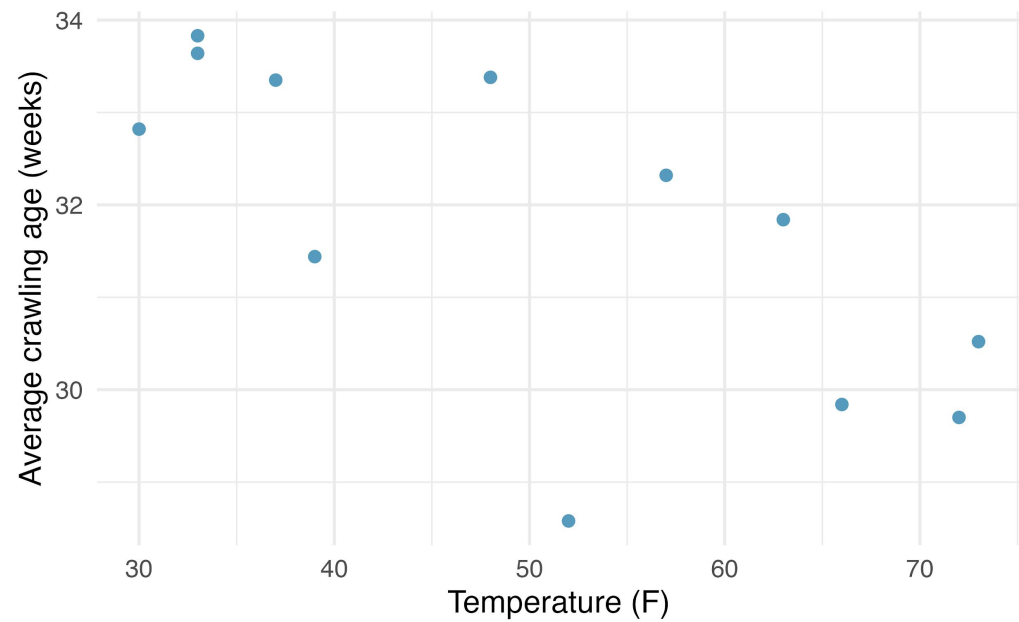
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Question 2

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Describe the relationship between average crawling age and average temperature as displayed in the scatterplot. Make sure to refer to the direction, strength and shape of the association, if any, and to speak in context.

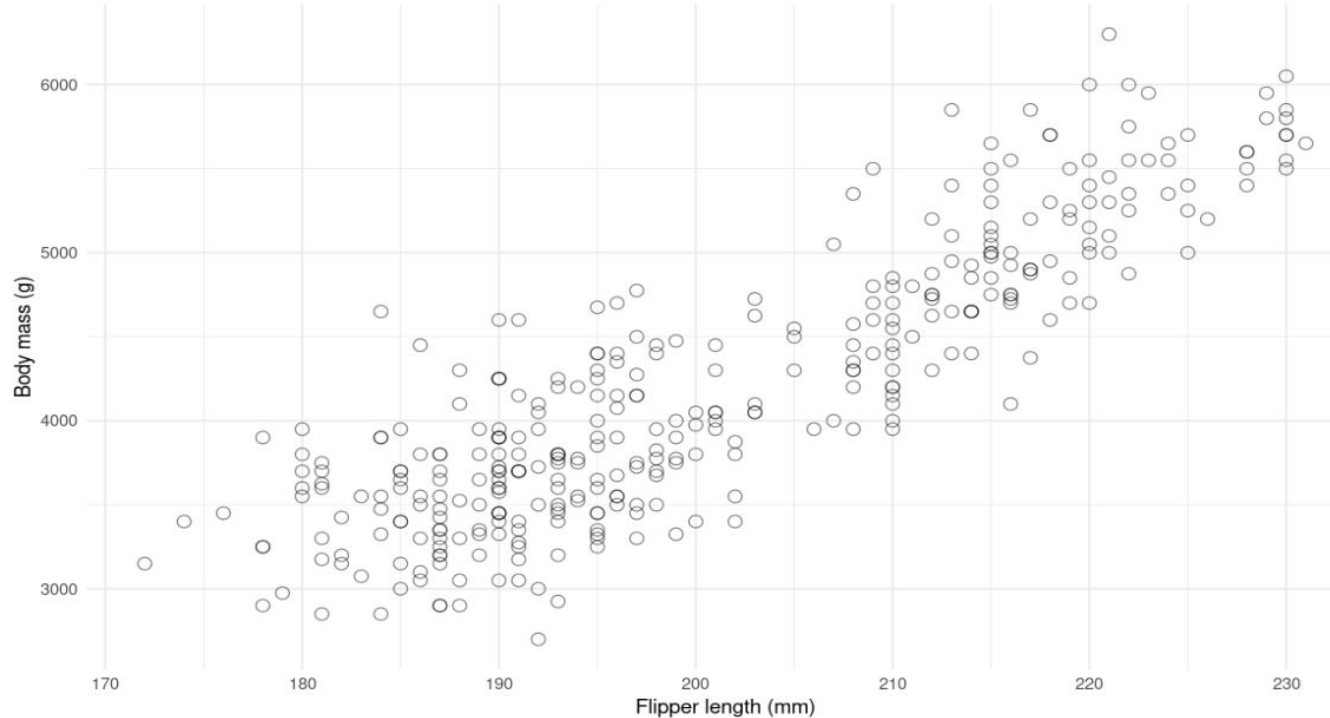


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Group 4 - Question 1

Researchers collected data on size measurements for three penguin species observed on three islands in the Palmer Archipelago, Antarctica. The data were collected from 2007-2009 as part of the US Long Term Ecological Research Network. Here is a graph of the penguins' body mass (in grams) versus their flipper length (in millimeters).

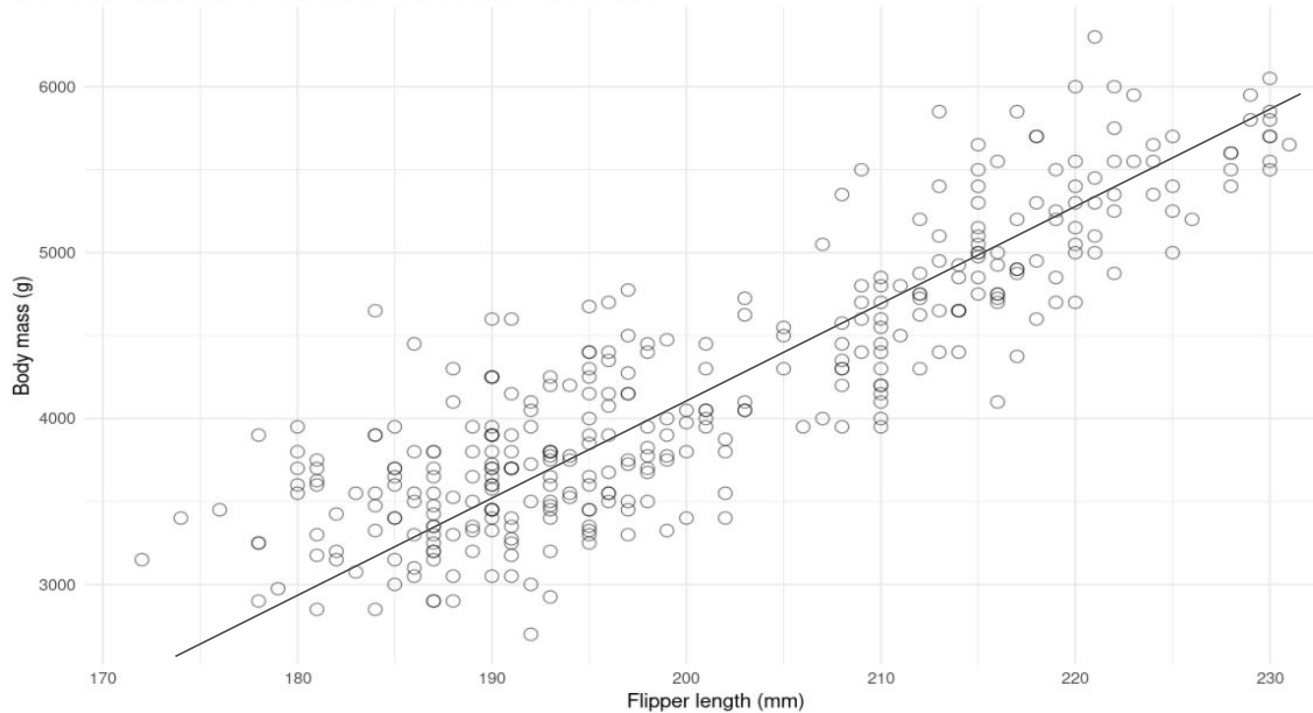
Penguin size, Palmer Station LTER
Flipper length and body mass for Adelie, Chinstrap and Gentoo Penguins



Go to the next slide to write your description.

Penguin size, Palmer Station LTER

Flipper length and body mass for Adelie, Chinstrap and Gentoo Penguins



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Describe the relationship between the penguins' body mass and their flipper length as displayed in the scatterplot. Make sure to refer to the direction, strength and shape of the association, if any, and to speak in context.

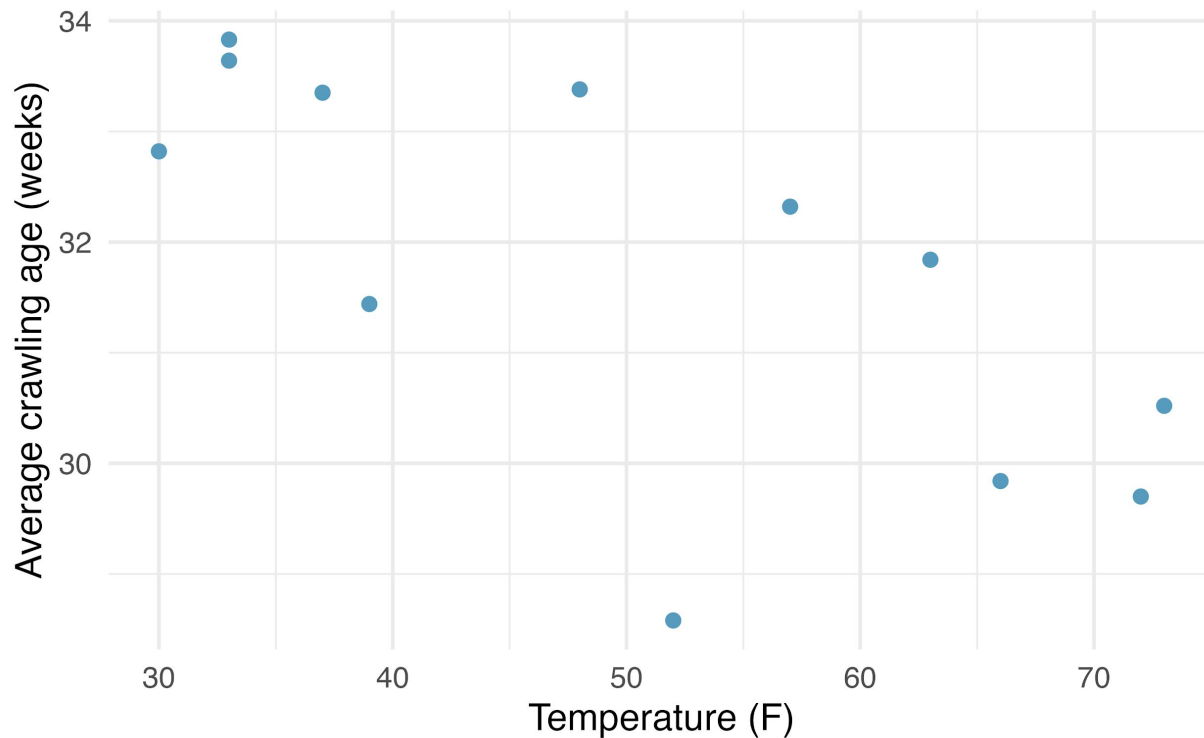
This is a graph of the relationship between penguin's flipper length (mm) and their body mass (g). These have a linear strong positive relationship: As flipper length increases penguins tend to have more body mass.

For example, one penguin had a flipper length of 185mm and weighed about 3000 grams, and another penguin had a flipper length of 230mm and weighed 5500 grams.

While this relationship is fairly strong, there still is some variation in weight for given flipper length values. For example, penguins with flippers at 190mm have weights that range between 3000g and around 4525g in observed cases.

Question 2

Crawling babies. A study conducted at the University of Denver investigated whether babies take longer to learn to crawl in cold months, when they are often bundled in clothes that restrict their movement, than in warmer months. Infants born during the study year were split into twelve groups, one for each birth month. We consider the average crawling age of babies in each group against the average temperature when the babies are six months old (that's when babies often begin trying to crawl). Temperature is measured in degrees Fahrenheit (F) and age is measured in weeks. (Benson 1993)

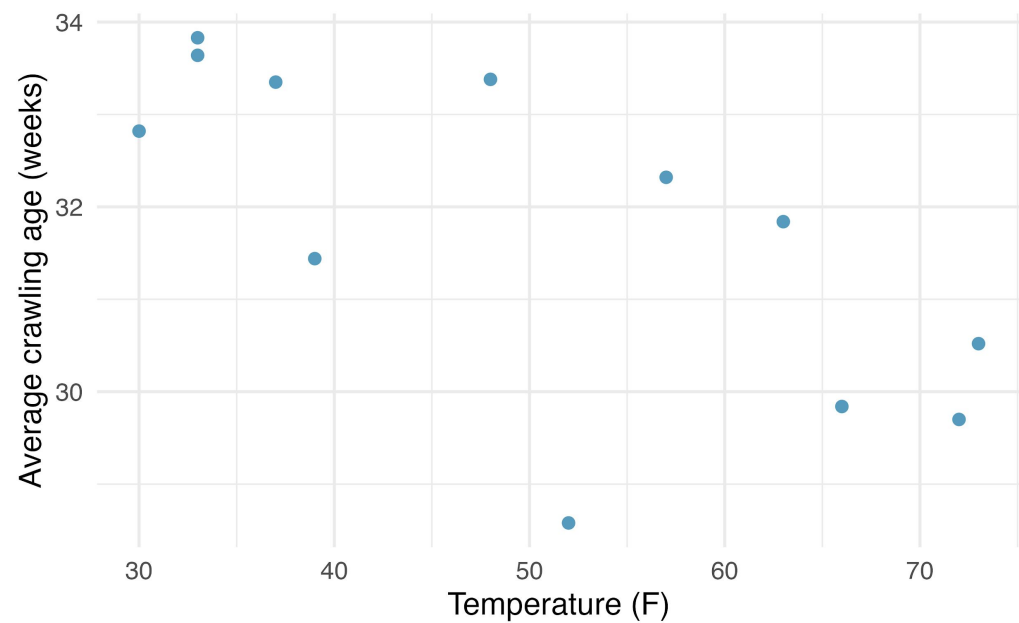


Describe the relationship between average crawling age and average temperature as displayed in the scatterplot. Make sure to refer to the direction, strength and shape of the association, if any, and to speak in context.

This is a graph of the relationship between the average temperature (F) when the each group was 6 months old and the average crawling age (weeks) of a group of babies. Generally, there is a linear, weak, negative association between the temperature and crawling age: As temperature increases, the babies tend to start crawling younger

For example, one baby was 6 months old when the temperature was averaging 32°F and started crawling at around 34 weeks, but another baby was 6 months old when the temperature was averaging 72°F and started crawling at around 30 weeks.

An exception to this trend is the group with an average temperature of approximately 52° F had an average crawling age of approximately 30.5 weeks.



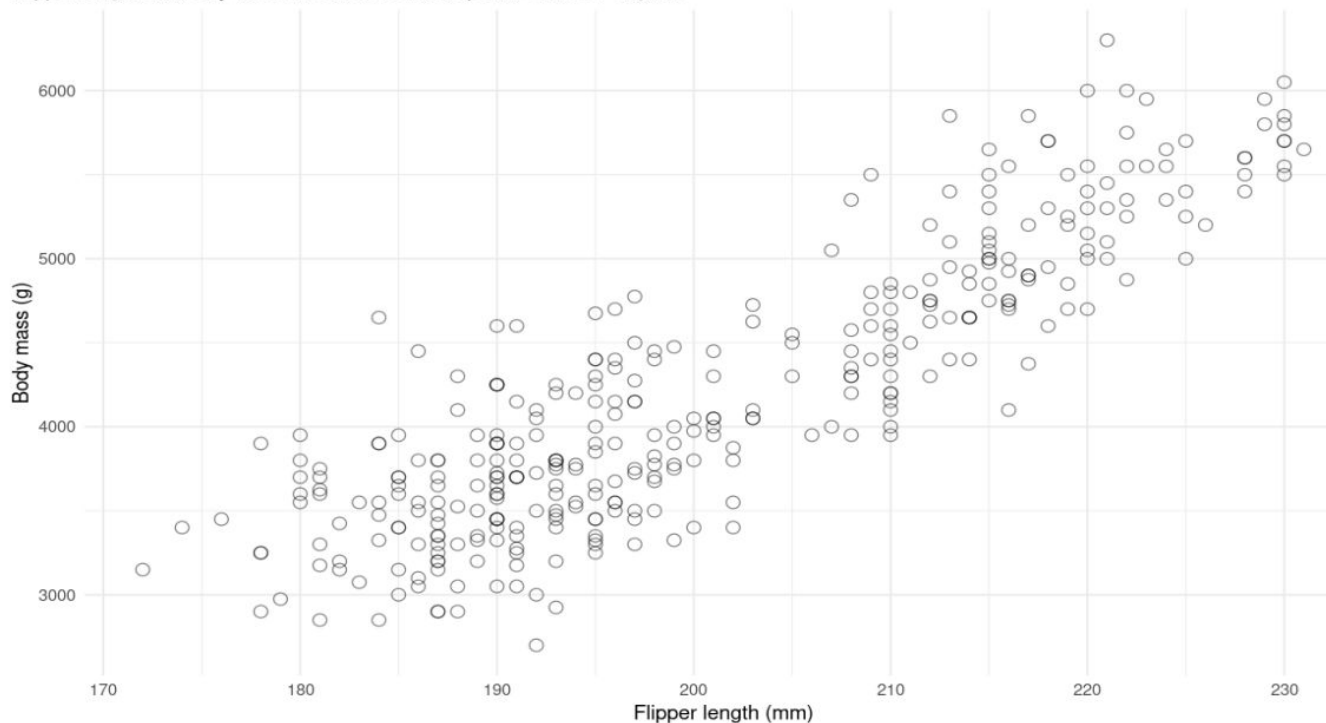
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Group 5 - Question 1

Researchers collected data on size measurements for three penguin species observed on three islands in the Palmer Archipelago, Antarctica. The data were collected from 2007-2009 as part of the US Long Term Ecological Research Network. Here is a graph of the penguins' body mass (in grams) versus their flipper length (in millimeters).

Penguin size, Palmer Station LTER

Flipper length and body mass for Adelie, Chinstrap and Gentoo Penguins



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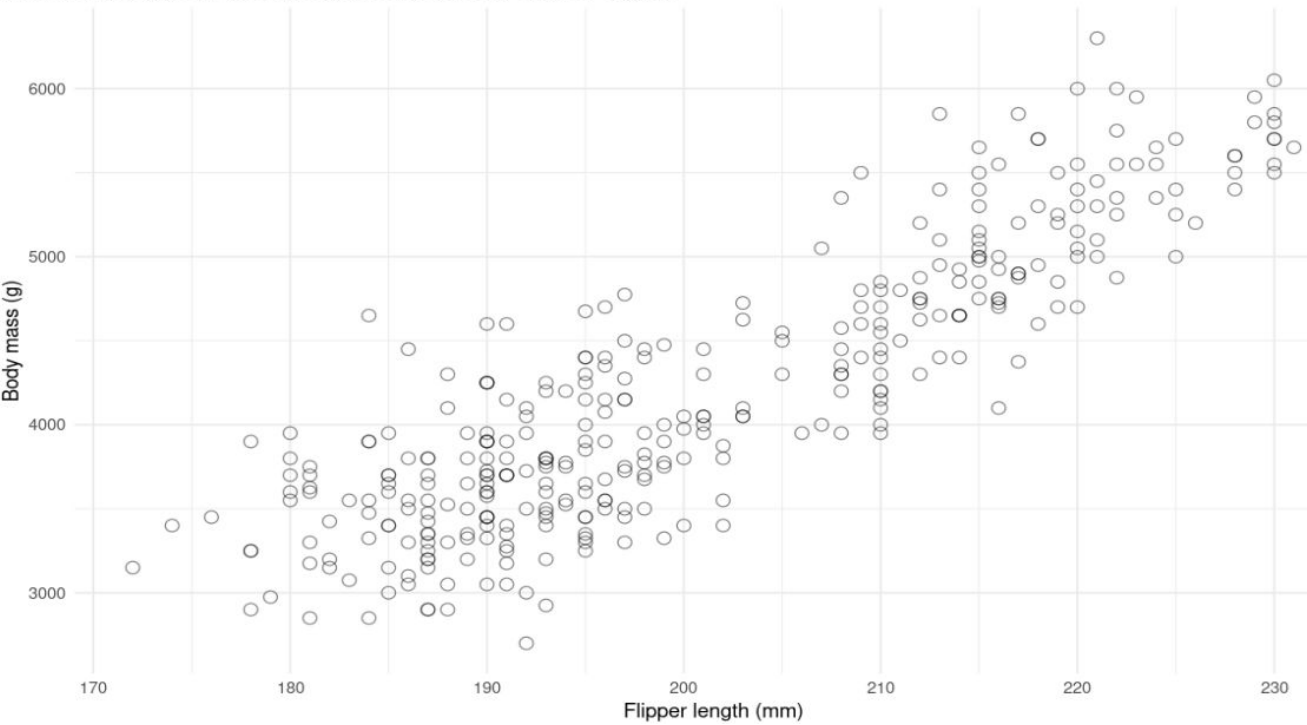
Describe the relationship between the penguins' body mass and their flipper length as displayed in the scatterplot. Make sure to refer to the direction, strength and shape of the association, if any, and to speak in context.

There is a strong, positive, linear relationship between penguins' body mass and flipper length; this suggests that the larger the body mass of the penguin, the longer the flipper will be.

For example, a penguin with flipper length of about 175 mm has a body mass of 3500g, while a penguin with flipper length 230 mm has a body mass of 6000g.

One outlier would be the penguin that seems to have a 193 flipper length, despite only being under 3000 g. One would think that it would have a shorter flipper with that little size, but it does not.

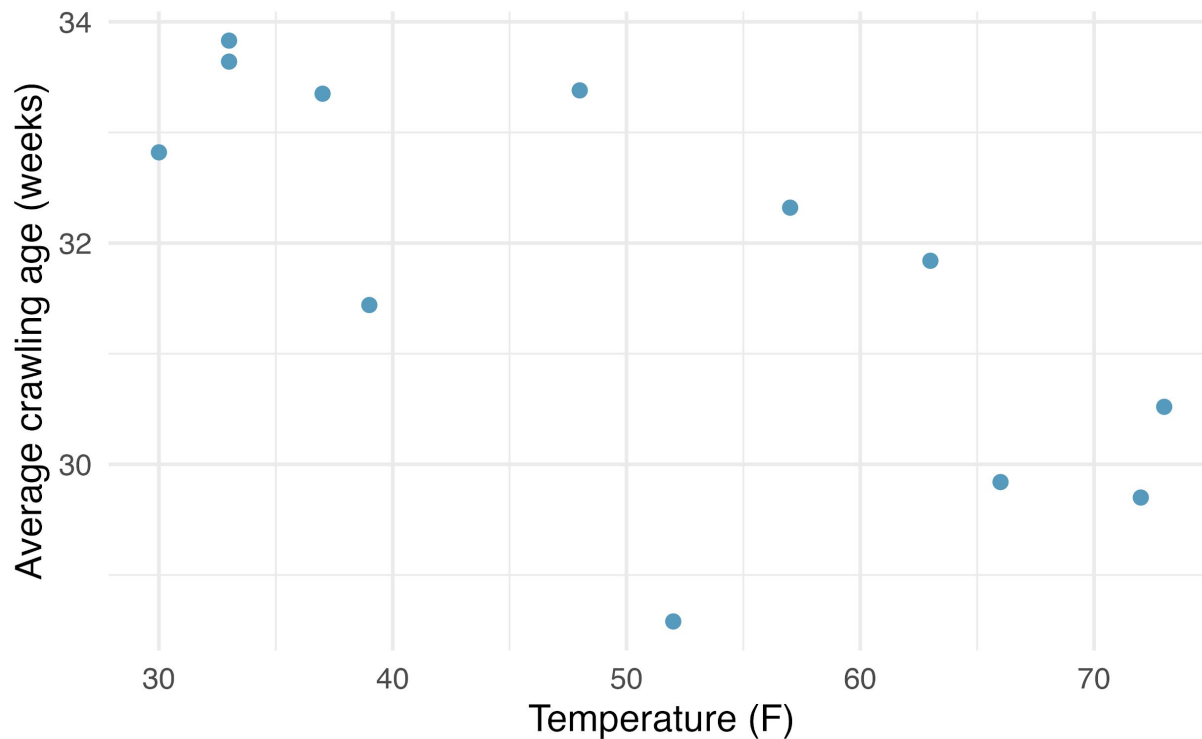
Penguin size, Palmer Station LTER
Flipper length and body mass for Adelie, Chinstrap and Gentoo Penguins



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Question 2

Crawling babies. A study conducted at the University of Denver investigated whether babies take longer to learn to crawl in cold months, when they are often bundled in clothes that restrict their movement, than in warmer months. Infants born during the study year were split into twelve groups, one for each birth month. We consider the average crawling age of babies in each group against the average temperature when the babies are six months old (that's when babies often begin trying to crawl). Temperature is measured in degrees Fahrenheit (F) and age is measured in weeks. (Benson 1993)

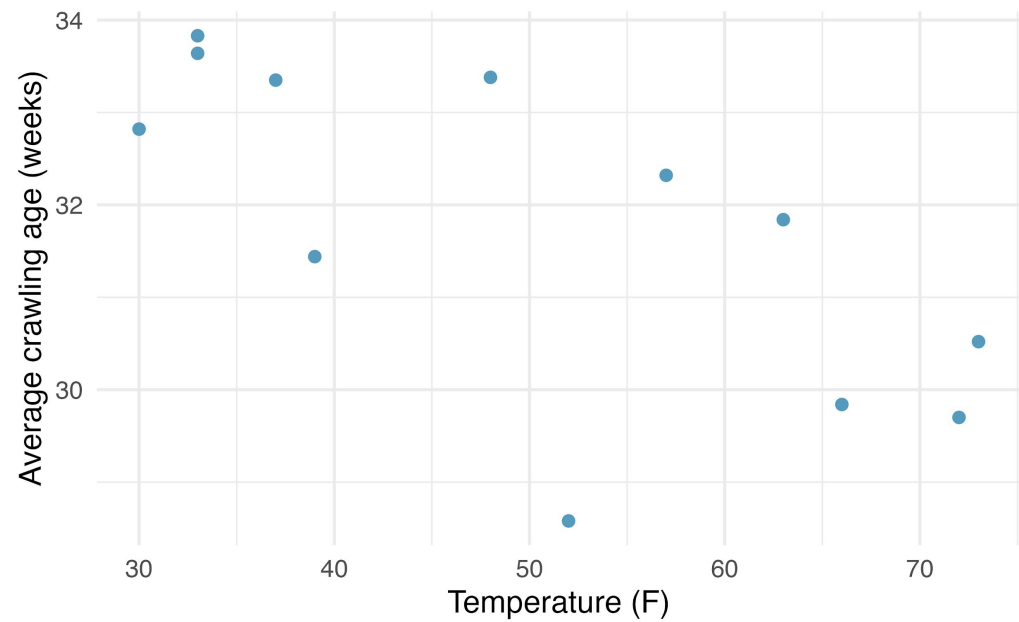


Describe the relationship between average crawling age and average temperature as displayed in the scatterplot. Make sure to refer to the direction, strength and shape of the association, if any, and to speak in context.

There is a moderate, negative, linear relationship between average crawling age and average temperature; this means that there is a greater average crawling age during the colder months and a lower average crawling age during warmer months.

For example, at temperature 33 degrees F, a baby took almost 34 weeks to learn how to crawl, but at temperature 72 degrees F, a baby only took a little less than 30 weeks to crawl.

One outlier to this would be the children who crawled at less than 29 weeks, despite it being around 53 degrees F. Being in the middle ranges, it should have taken these children at least weeks, but they managed to do it faster than all the others.



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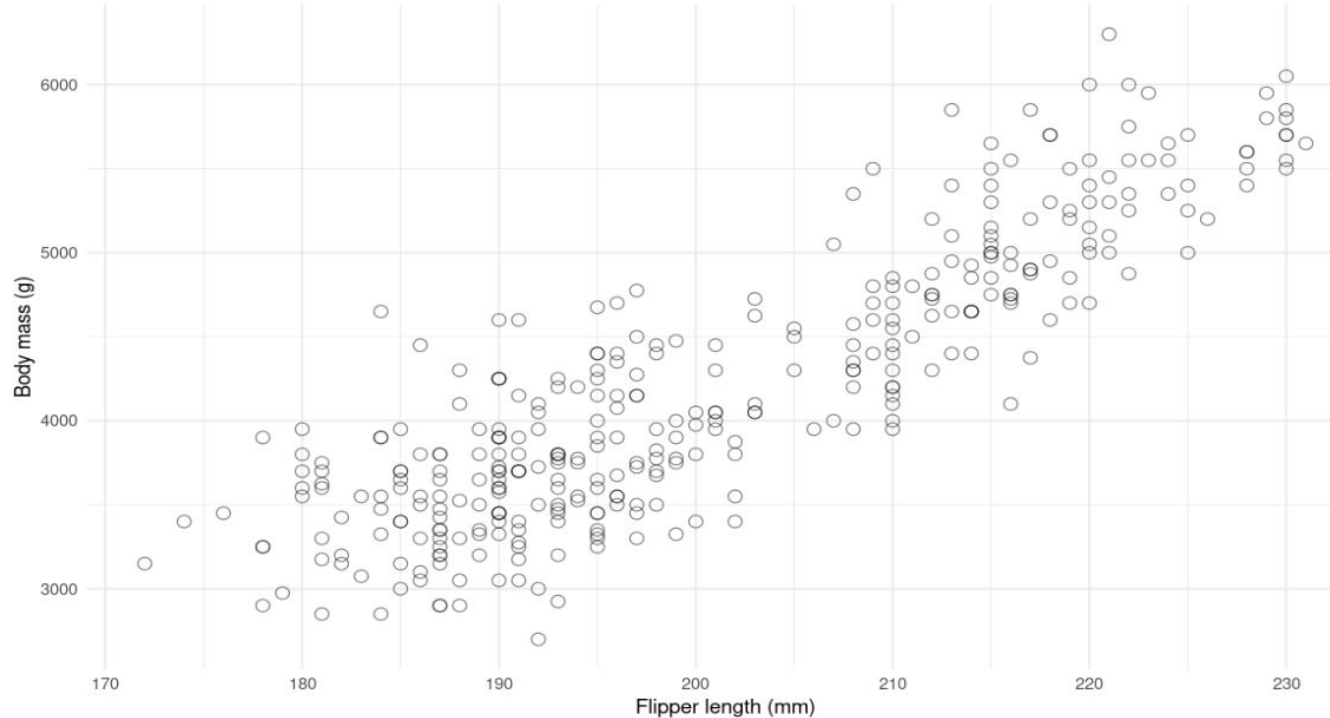
Group 6 - Question 1

Researchers collected data on size measurements for three penguin species observed on three islands in the Palmer Archipelago, Antarctica. The data were collected from 2007-2009 as part of the US Long Term Ecological Research Network. Here is a graph of the penguins' body mass (in grams) versus their flipper length (in millimeters).

Go to the next slide to write your description.

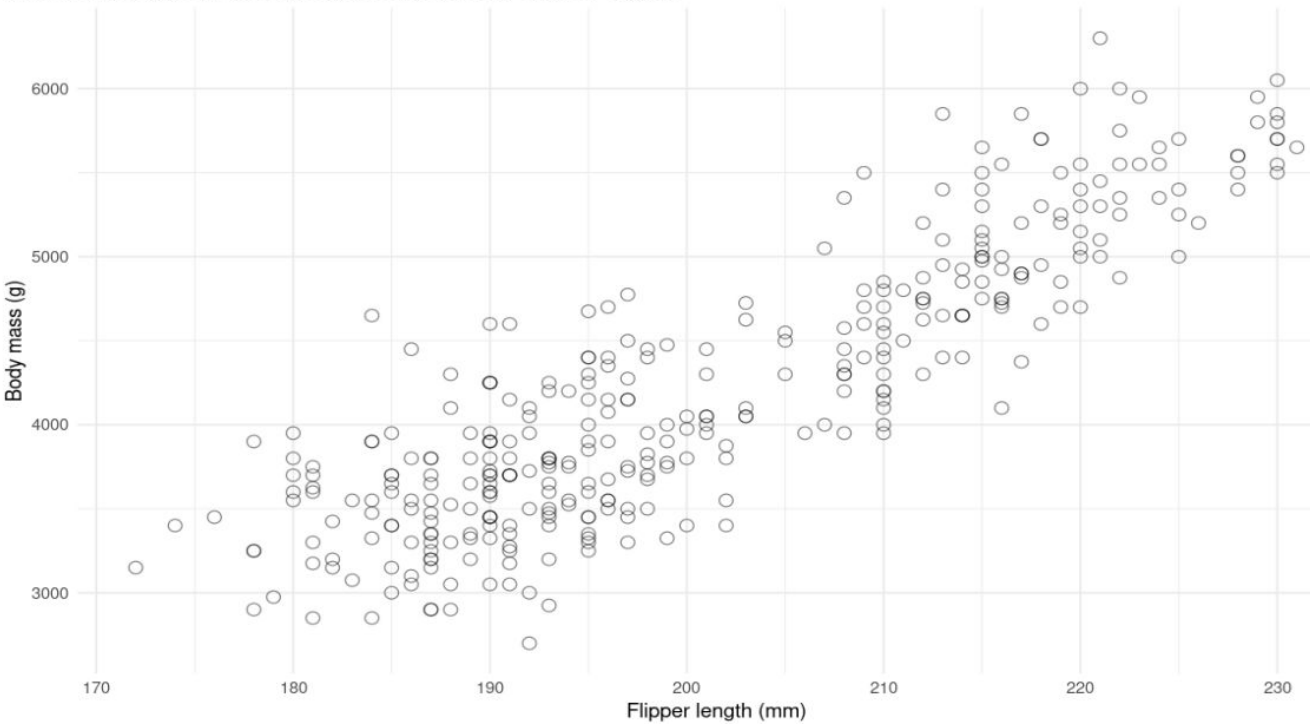
Penguin size, Palmer Station LTER

Flipper length and body mass for Adelie, Chinstrap and Gentoo Penguins



Describe the relationship between the penguins' body mass and their flipper length as displayed in the scatterplot. Make sure to refer to the direction, strength and shape of the association, if any, and to speak in context.

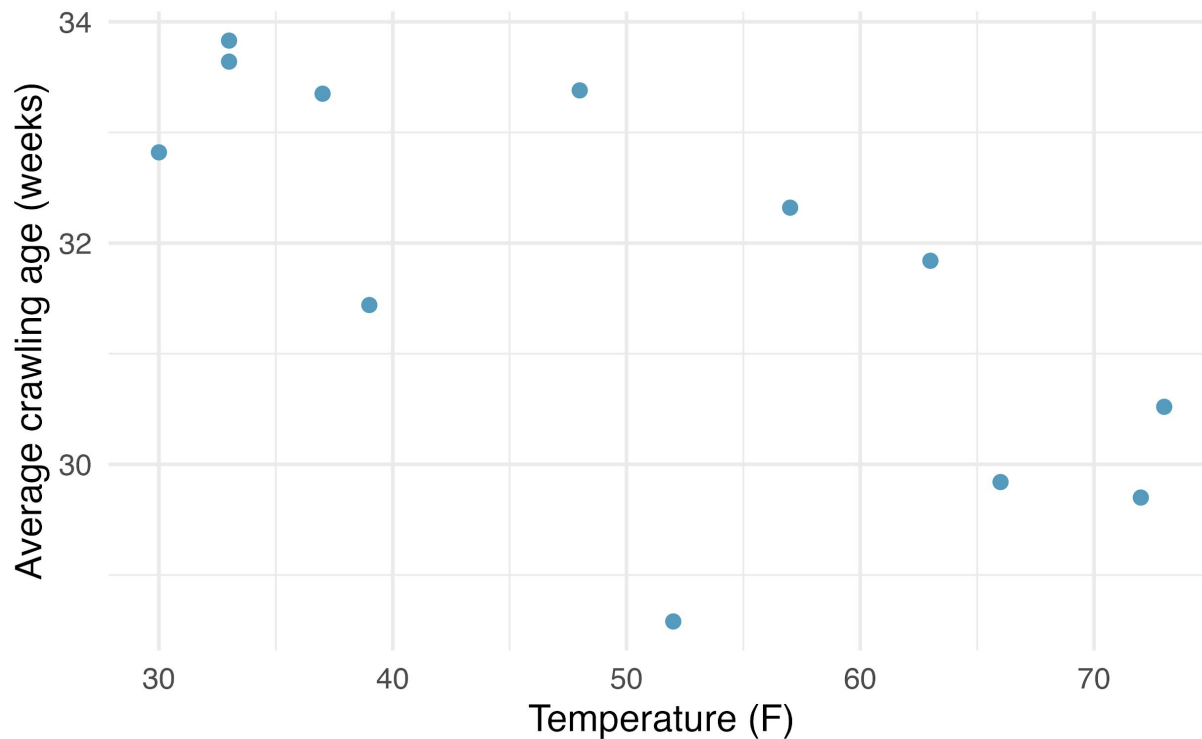
Penguin size, Palmer Station LTER
Flipper length and body mass for Adelie, Chinstrap and Gentoo Penguins



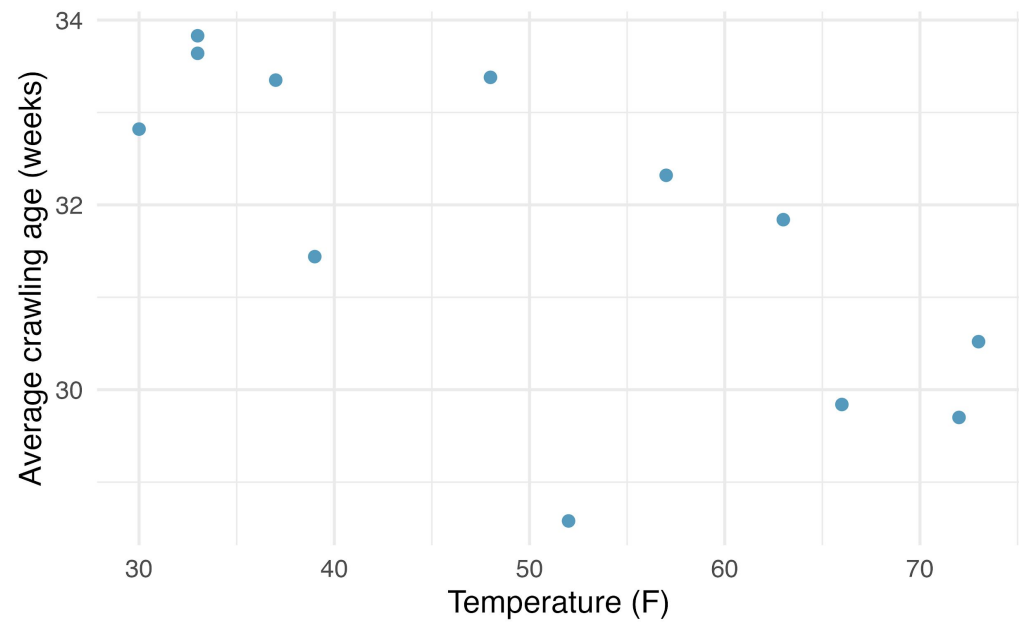
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Question 2

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Describe the relationship between average crawling age and average temperature as displayed in the scatterplot. Make sure to refer to the direction, strength and shape of the association, if any, and to speak in context.

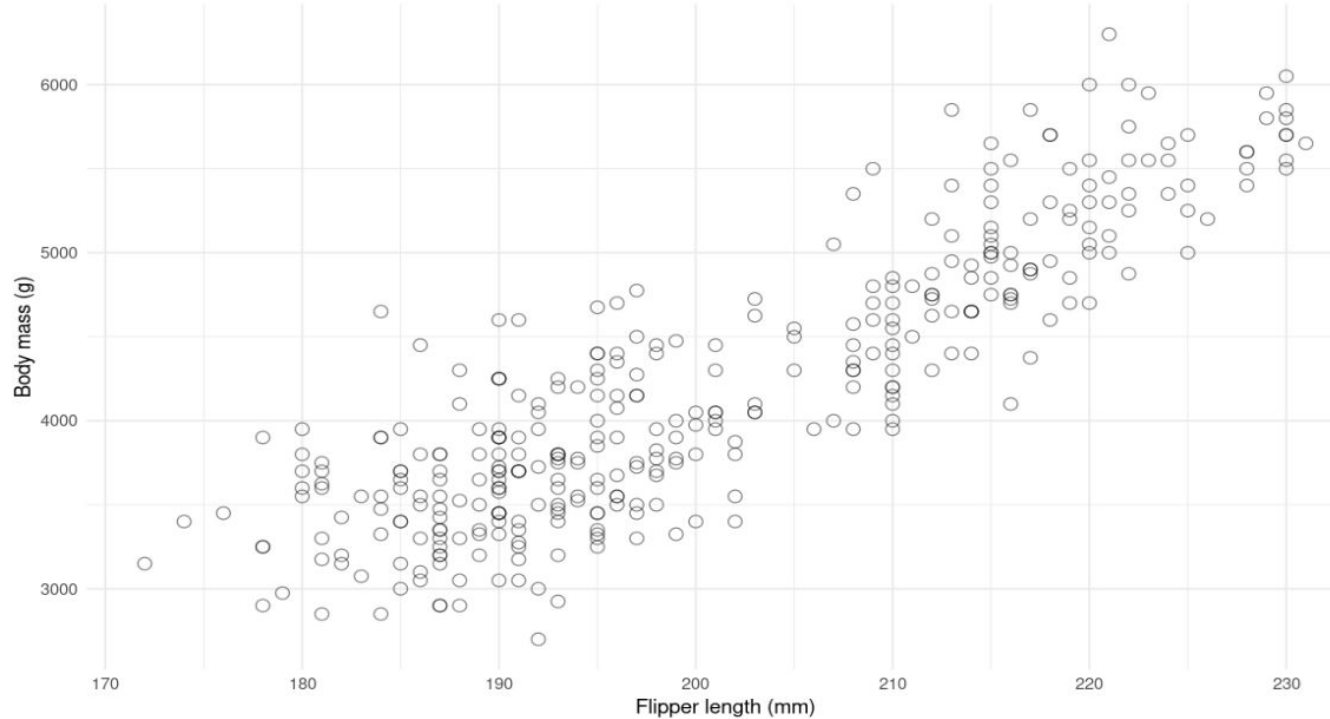


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Group 7 - Question 1

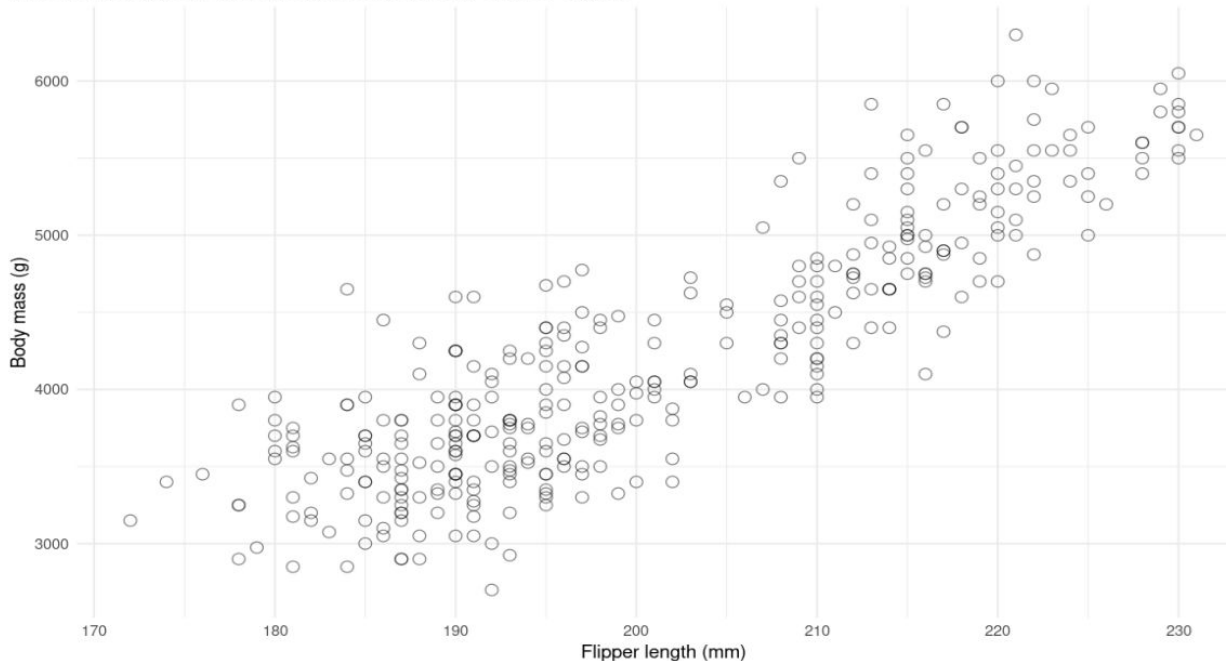
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Penguin size, Palmer Station LTER
Flipper length and body mass for Adelie, Chinstrap and Gentoo Penguins



Go to the next slide to write your description.

Penguin size, Palmer Station LTER
Flipper length and body mass for Adelie, Chinstrap and Gentoo Penguins



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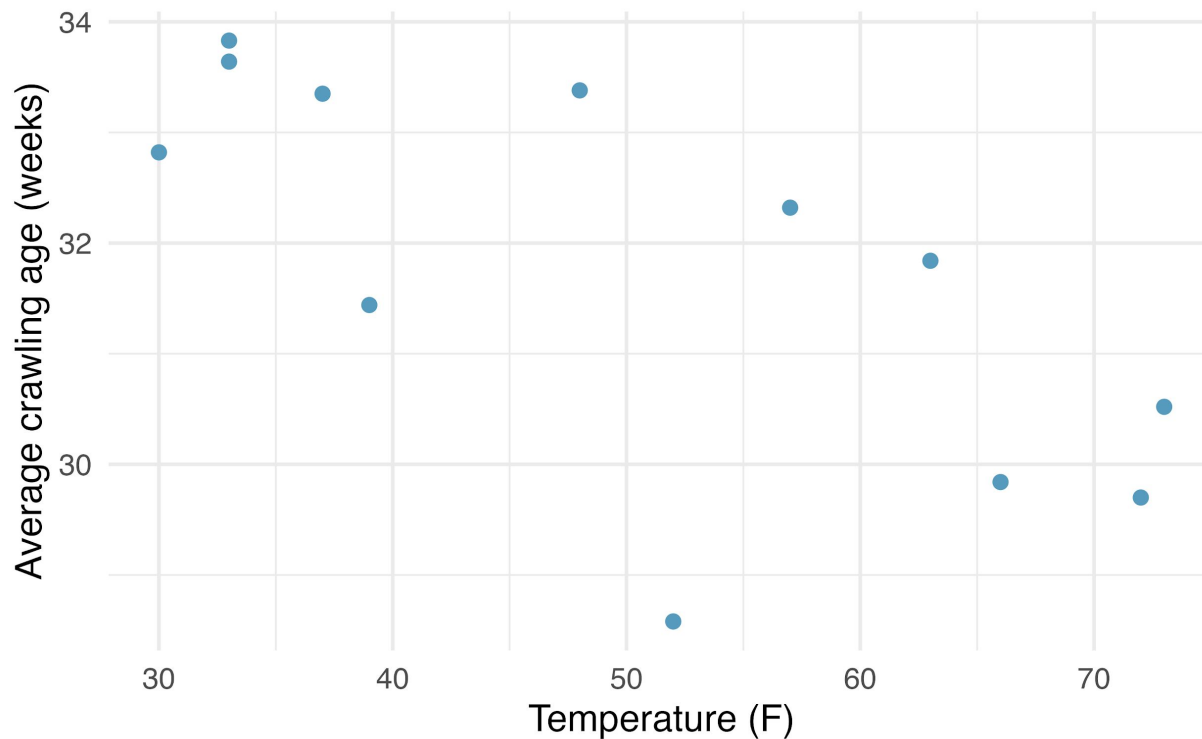
Describe the relationship between the penguins' body mass and their flipper length as displayed in the scatterplot. Make sure to refer to the direction, strength and shape of the association, if any, and to speak in context.

This scatterplot shows the relationship between the body mass and flipper length. There is a positive linear association between these variables penguins with a larger body mass tend to have longer flipper length.

For example one penguin has a body mass of 4000 g and a flipper length of 190 mm and there is another with a body mass of 4400 g and a flipper length of 197 mm. However there is some variation in the flipper length and body mass for example one penguin has a flipper length of 183 mm and a body mass of 4600 g.

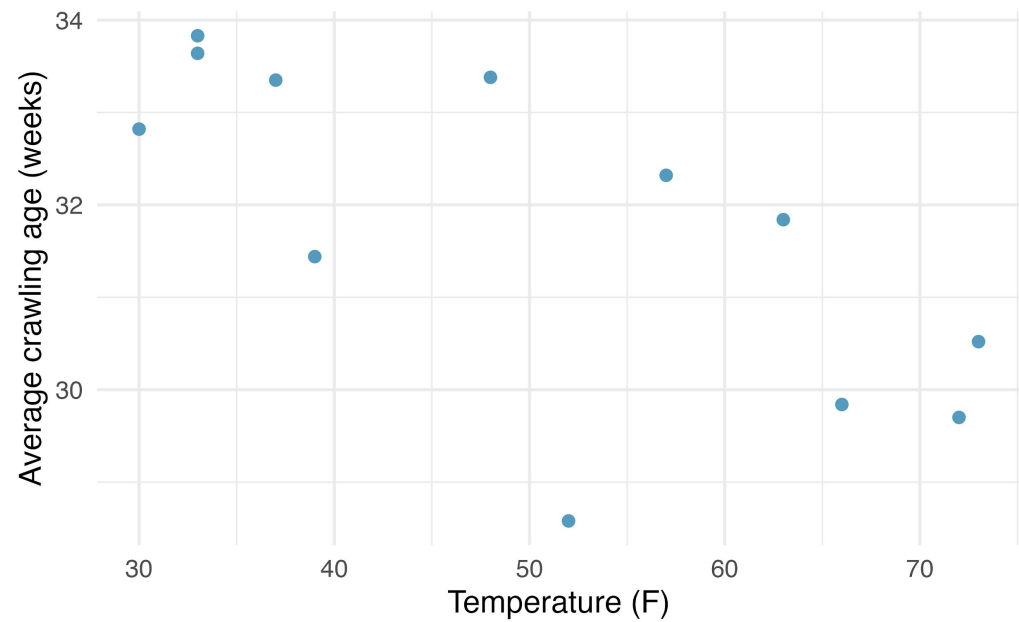
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Describe the relationship between average crawling age and average temperature as displayed in the scatterplot. Make sure to refer to the direction, strength and shape of the association, if any, and to speak in context.

The scatterplot shows the relationship between average temperature of kids and the average crawling age. Generally, it has a linear weak negative association, colder temperatures will increase the average crawling age, and warmer temperatures have the baby crawling sooner. For example, if the average temp is about 35 degrees Fahrenheit, and another has an average temp of 66 degrees Fahrenheit and an average crawling age of 29 weeks. The relationship is only weak, meaning there is a small difference between average crawling age and temperature.



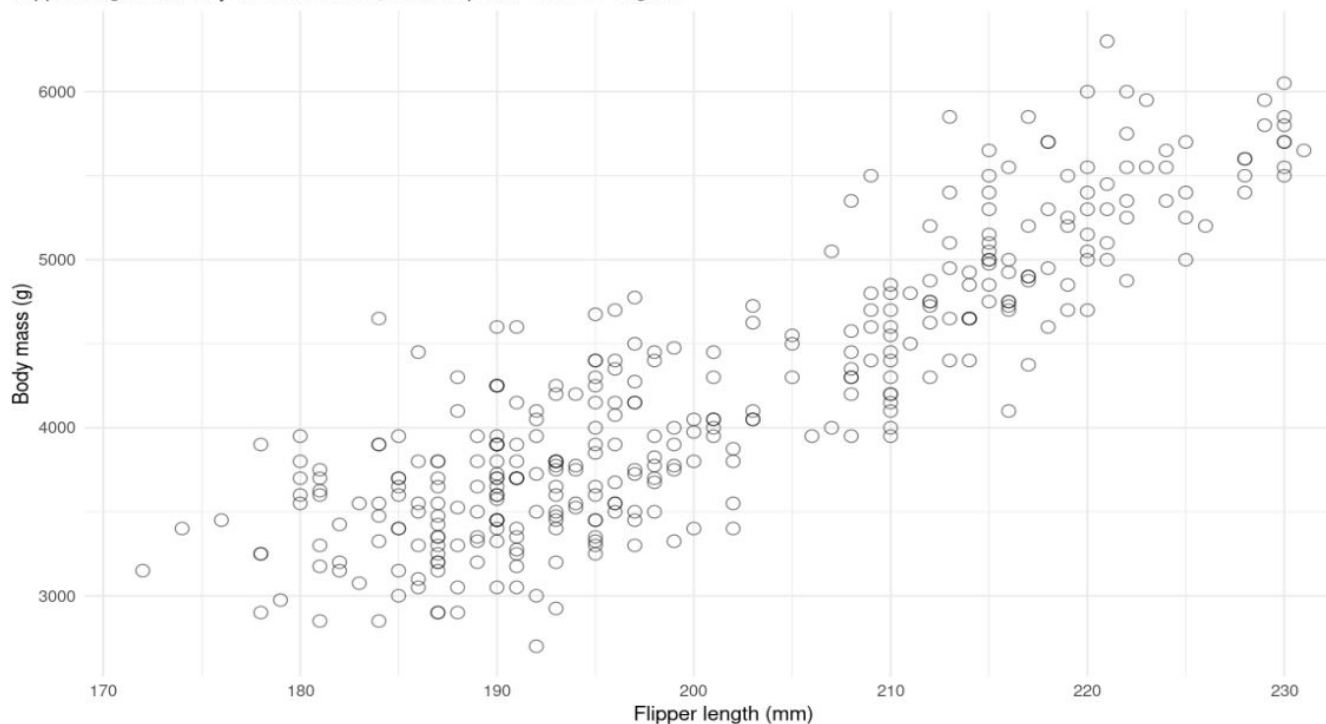
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Group 8 - Question 1

Researchers collected data on size measurements for three penguin species observed on three islands in the Palmer Archipelago, Antarctica. The data were collected from 2007-2009 as part of the US Long Term Ecological Research Network. Here is a graph of the penguins' body mass (in grams) versus their flipper length (in millimeters).

Penguin size, Palmer Station LTER

Flipper length and body mass for Adelie, Chinstrap and Gentoo Penguins

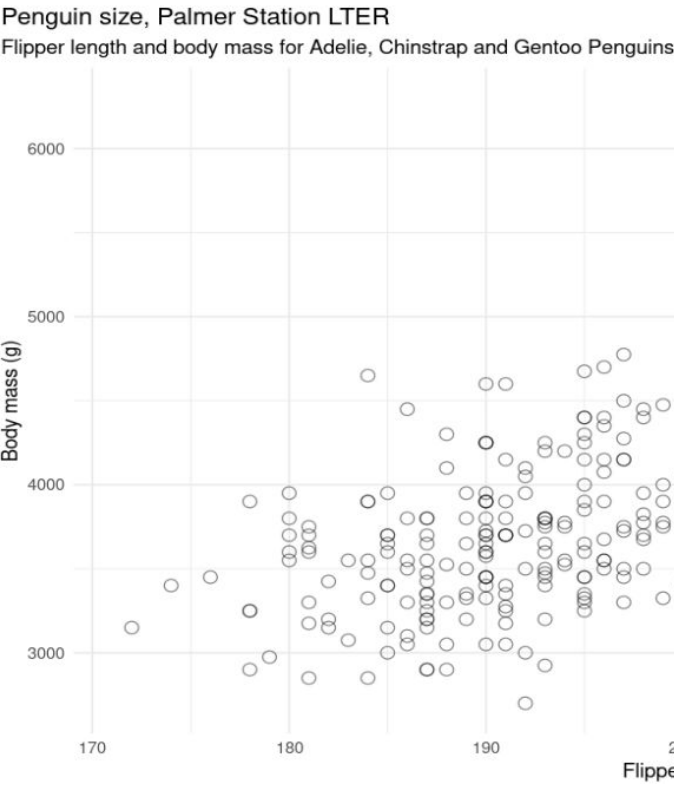


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Describe the relationship between the penguins' body mass and their flipper length as displayed in the scatterplot. Make sure to refer to the direction, strength and shape of the association, if any, and to speak in context.

The graph is a depiction of body mass vs. their flipper length of Adelie, Chinstrap, and Gentoo Penguins. The graph shows a strong, positive, linear association between the variables: the greater the flipper length (mm) of the penguins, the greater their body mass (g).

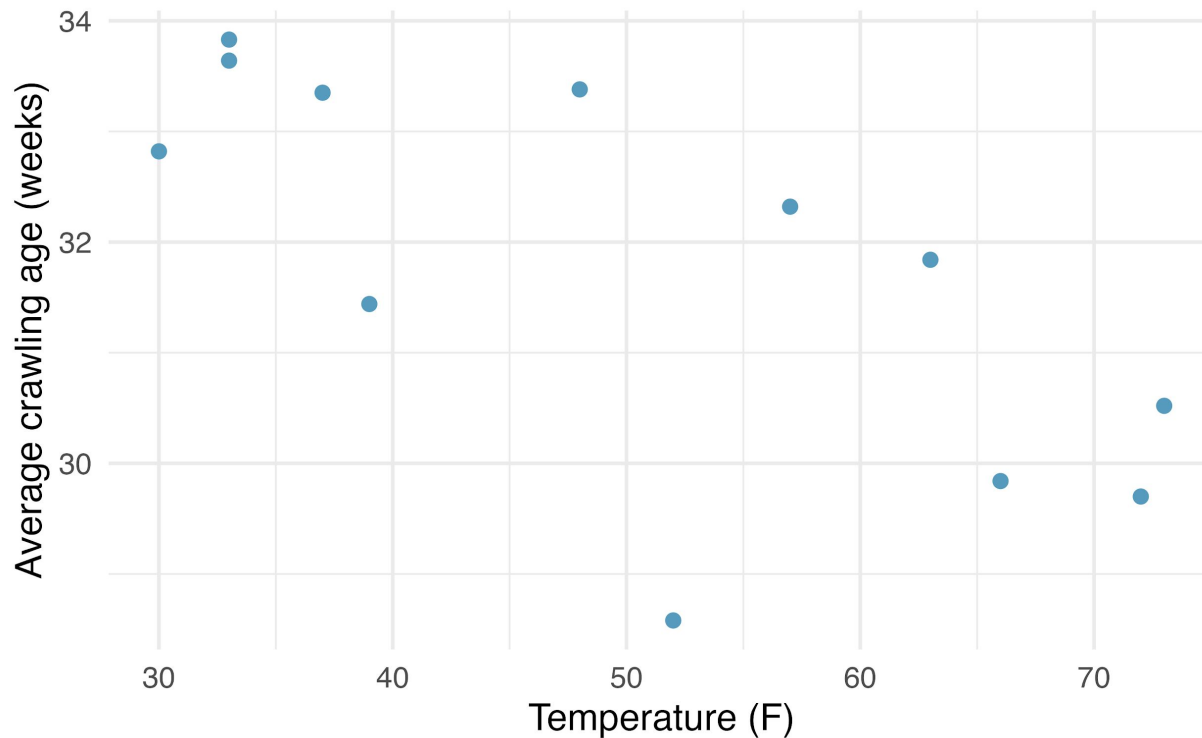
For example, penguins with a flipper length of 180mm weighs in a range between 3550g and 3950g, and penguins with a flipper length of 230mm weighs in a range between 5500g and 5800g.



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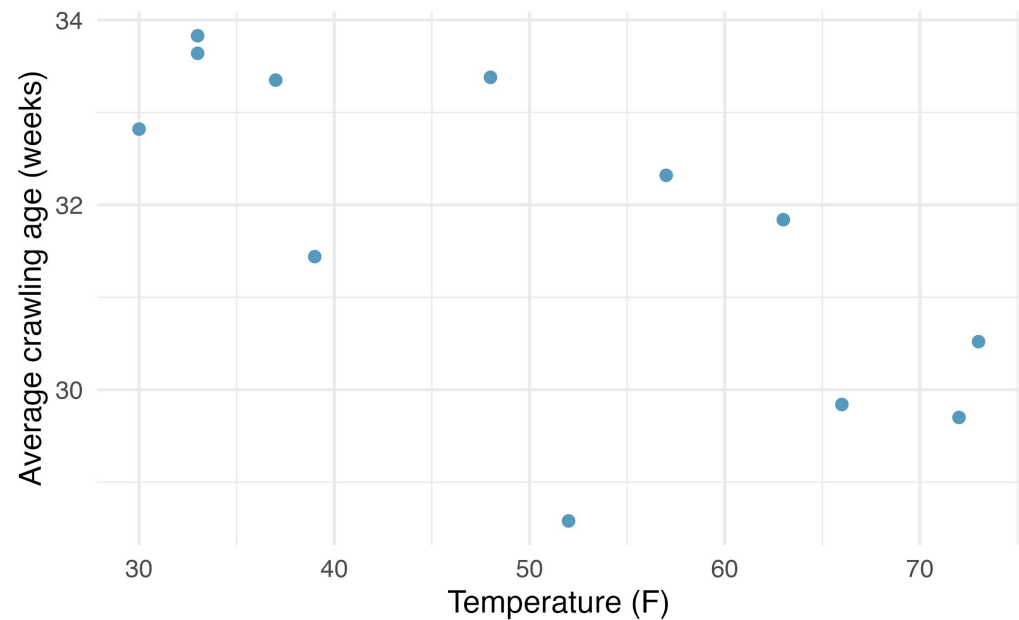
Question 2

Crawling babies. A study conducted at the University of Denver investigated whether babies take longer to learn to crawl in cold months, when they are often bundled in clothes that restrict their movement, than in warmer months. Infants born during the study year were split into twelve groups, one for each birth month. We consider the average crawling age of babies in each group against the average temperature when the babies are six months old (that's when babies often begin trying to crawl). Temperature is measured in degrees Fahrenheit (F) and age is measured in weeks. (Benson 1993)



Describe the relationship between average crawling age and average temperature as displayed in the scatterplot. Make sure to refer to the direction, strength and shape of the association, if any, and to speak in context.

The scatter plot depicts the relationship between an infant's age at the time they begin crawling with the average temperature at their 6mo age. There is a moderate negative linear association between these variables: babies in colder temperatures are more likely to be delayed in their crawling. For example: Babies with an average temp of 30° (F) at their 6mo point began crawling late in their 32nd week and those with an average temp in the 70°s (F) began crawling as early as their 30th week. There is only a moderately strong relationship meaning there is some variation in the temperature/crawling age relationship and some exceptions do exists such as the baby that was crawling at 31.5 week despite the sub 40°(F) temperatures.



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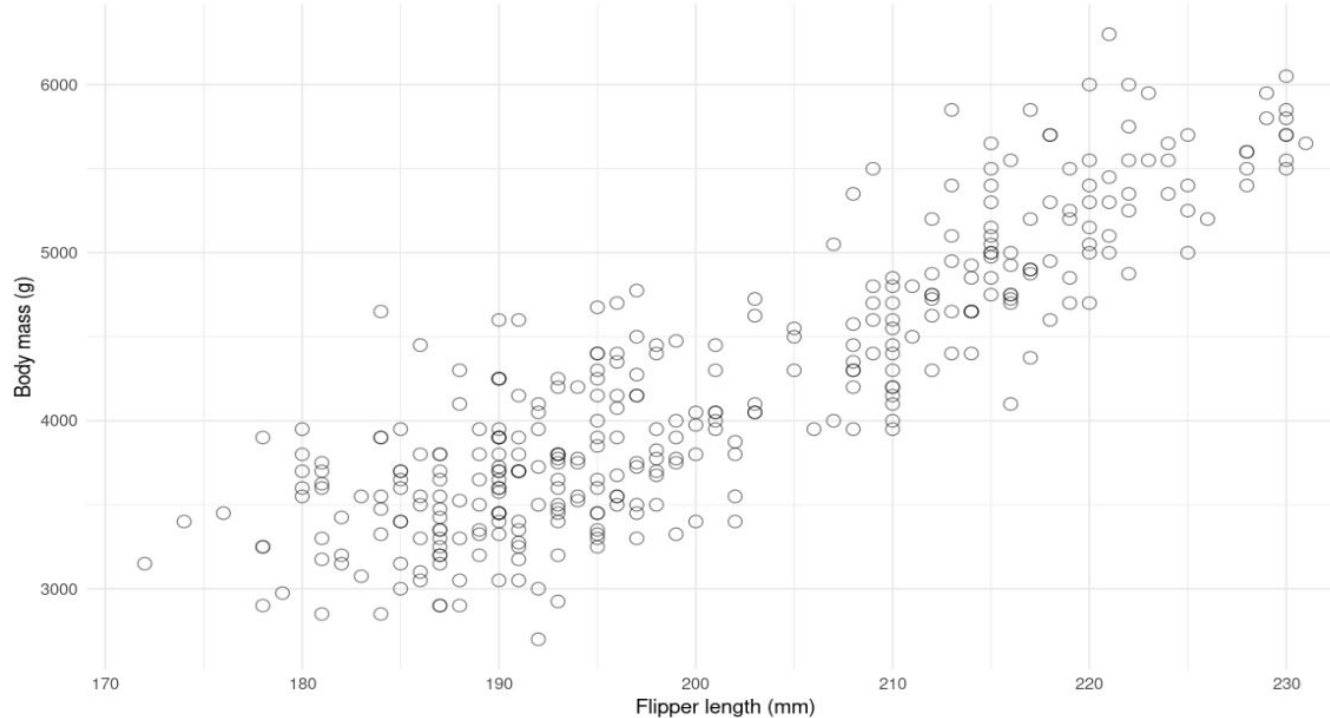
Group 9 - Question 1

Researchers collected data on size measurements for three penguin species observed on three islands in the Palmer Archipelago, Antarctica. The data were collected from 2007-2009 as part of the US Long Term Ecological Research Network. Here is a graph of the penguins' body mass (in grams) versus their flipper length (in millimeters).

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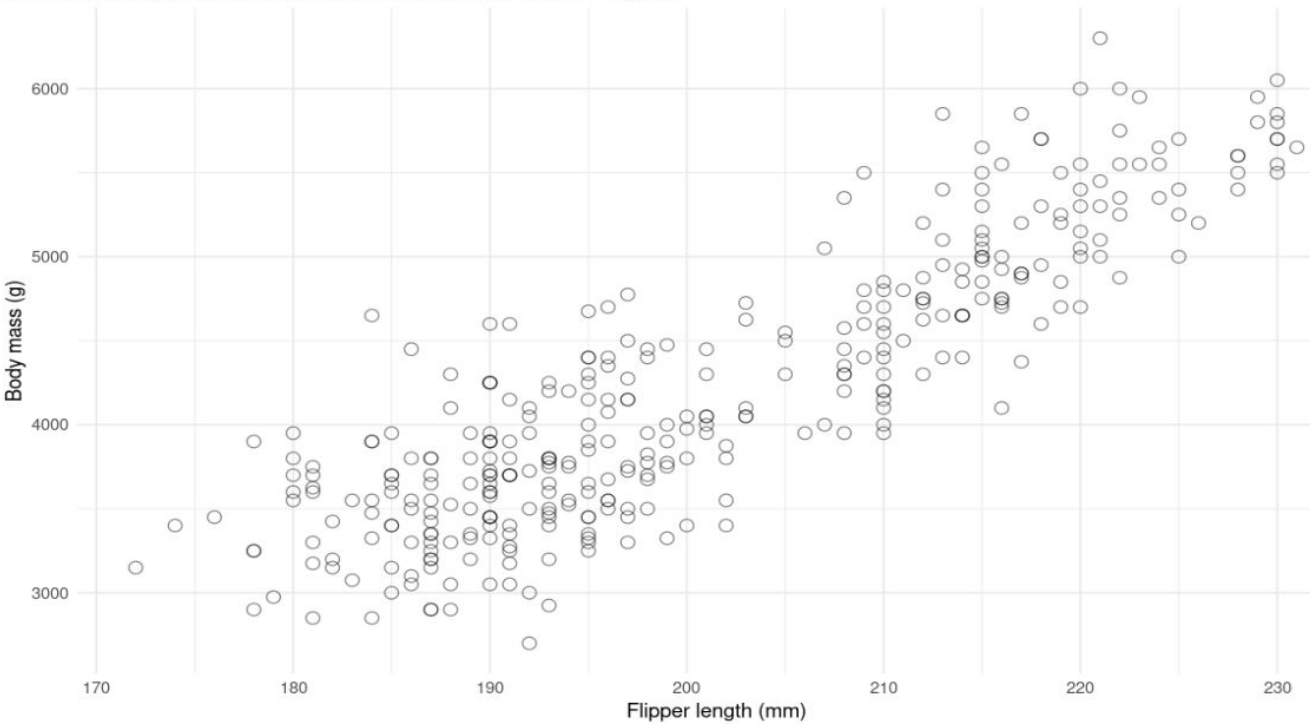
Penguin size, Palmer Station LTER

Flipper length and body mass for Adelie, Chinstrap and Gentoo Penguins



Describe the relationship between the penguins' body mass and their flipper length as displayed in the scatterplot. Make sure to refer to the direction, strength and shape of the association, if any, and to speak in context.

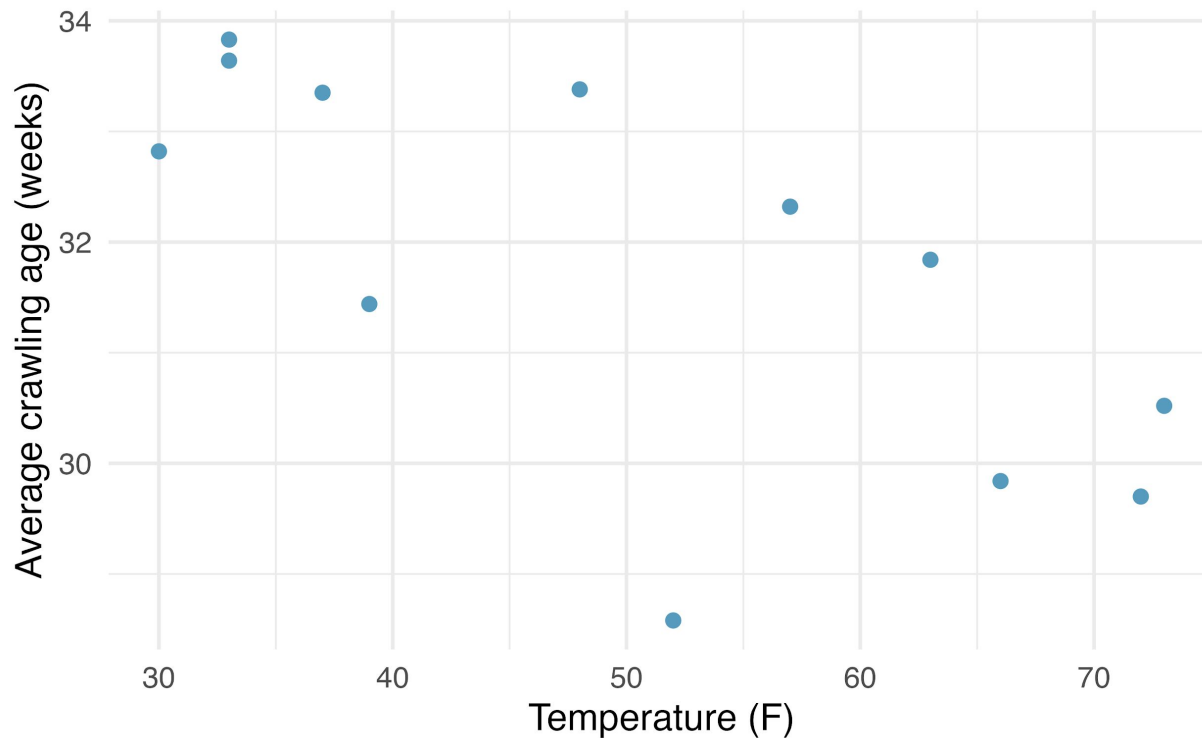
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Flipper length and body mass for Adelie, Chinstrap and Gentoo Penguins



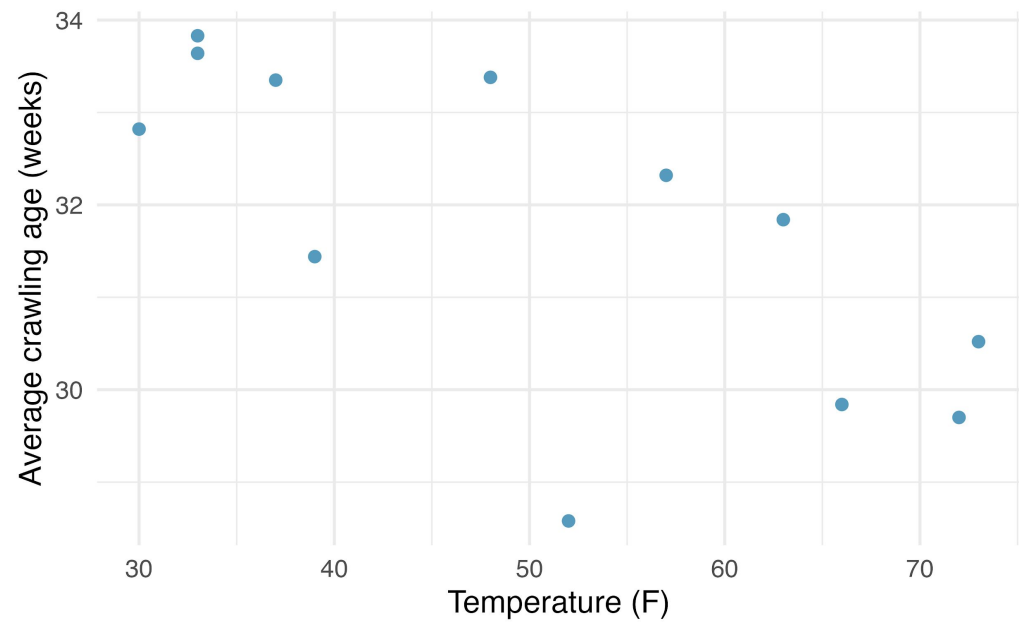
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Describe the relationship between average crawling age and average temperature as displayed in the scatterplot. Make sure to refer to the direction, strength and shape of the association, if any, and to speak in context.

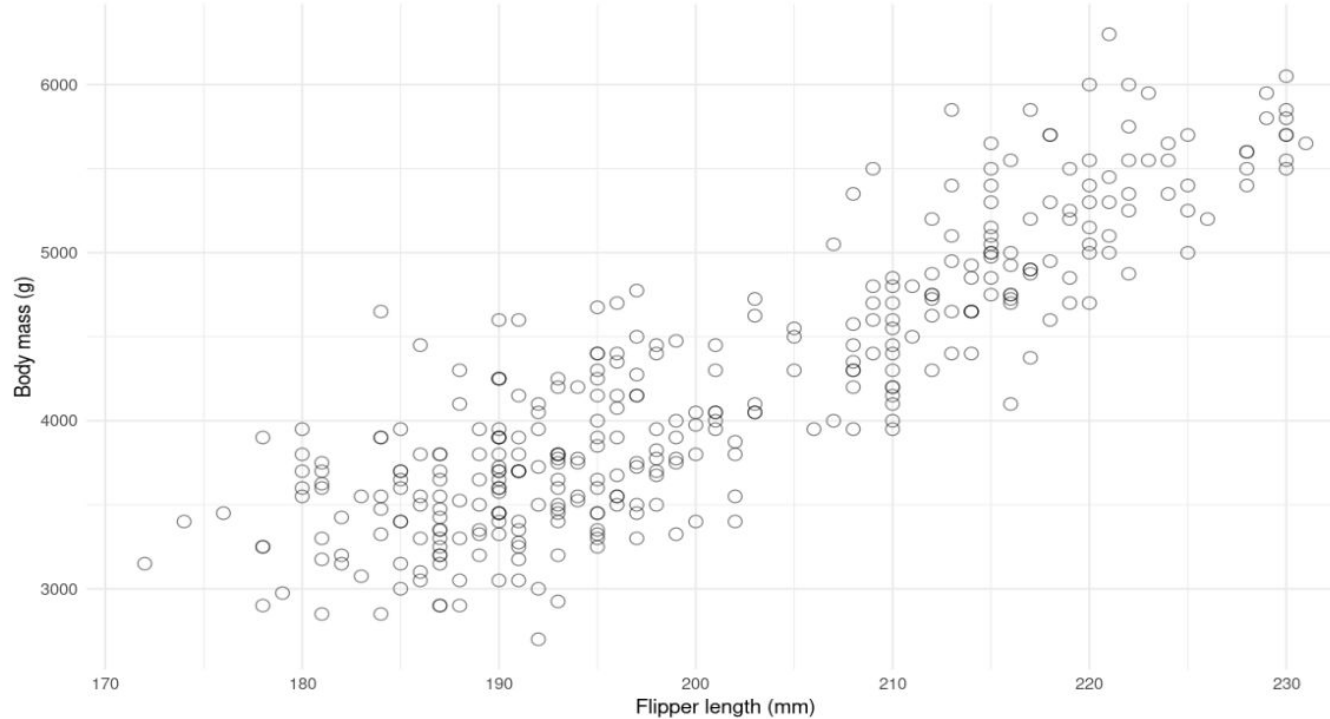


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Group 10 - Question 1

Researchers collected data on size measurements for three penguin species observed on three islands in the Palmer Archipelago, Antarctica. The data were collected from 2007-2009 as part of the US Long Term Ecological Research Network. Here is a graph of the penguins' body mass (in grams) versus their flipper length (in millimeters).

Penguin size, Palmer Station LTER
Flipper length and body mass for Adelie, Chinstrap and Gentoo Penguins

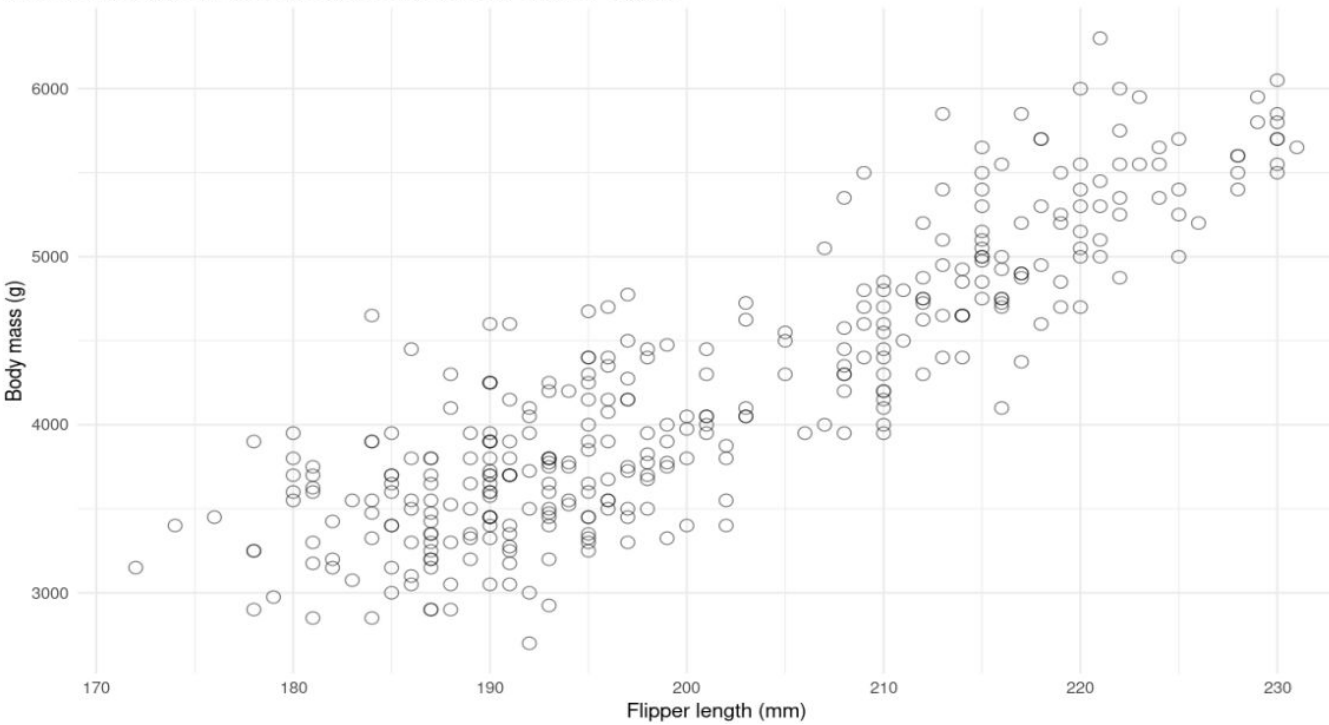


Go to the next slide to write your description.

Describe the relationship between the penguins' body mass and their flipper length as displayed in the scatterplot. Make sure to refer to the direction, strength and shape of the association, if any, and to speak in context.

The scatterplot shows the relationship between the length of a penguins flipper and their body mass. Generally, there is a positive linear trend between these two variables. The longer the penguins flipper length the heavier the body mass of the penguin. For example, one penguins body mass is 3950 grams and its flipper length is 180 mm, while another penguins body mass is 6000 grams and its flipper length is 220 mm. The relationship is moderately strong, which means there is a relationship between body mass and flipper length in penguins.

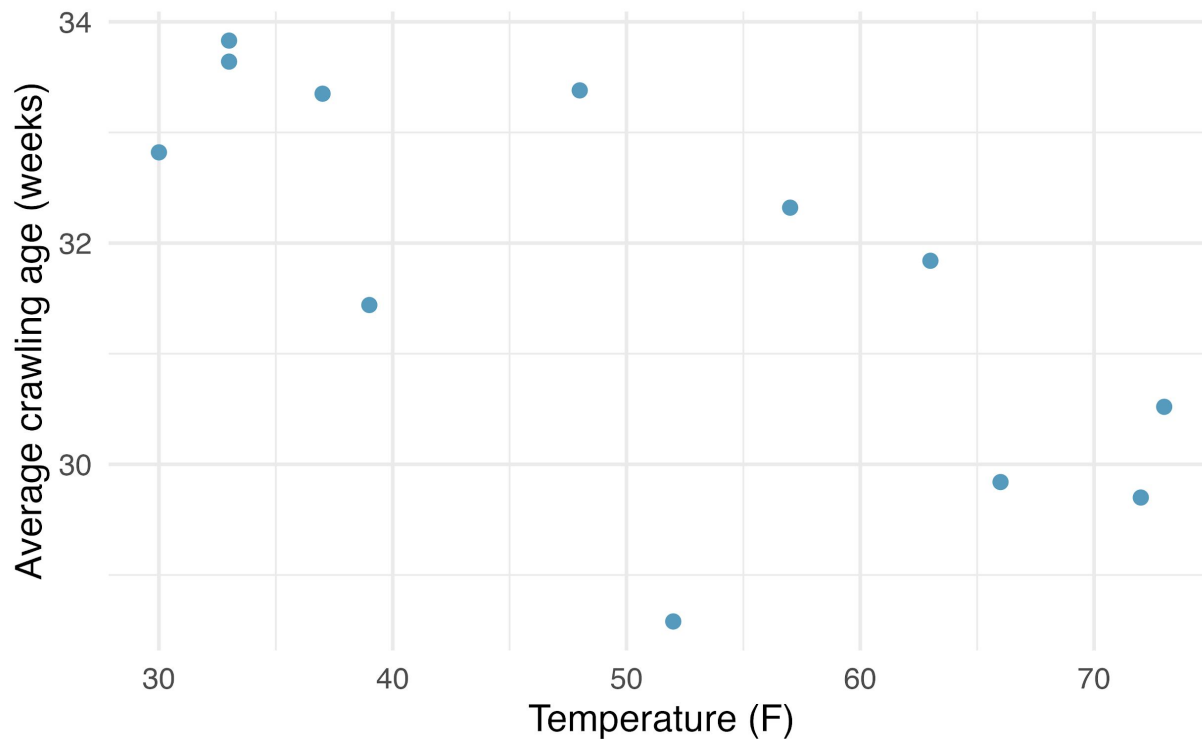
Penguin size, Palmer Station LTER
Flipper length and body mass for Adelie, Chinstrap and Gentoo Penguins



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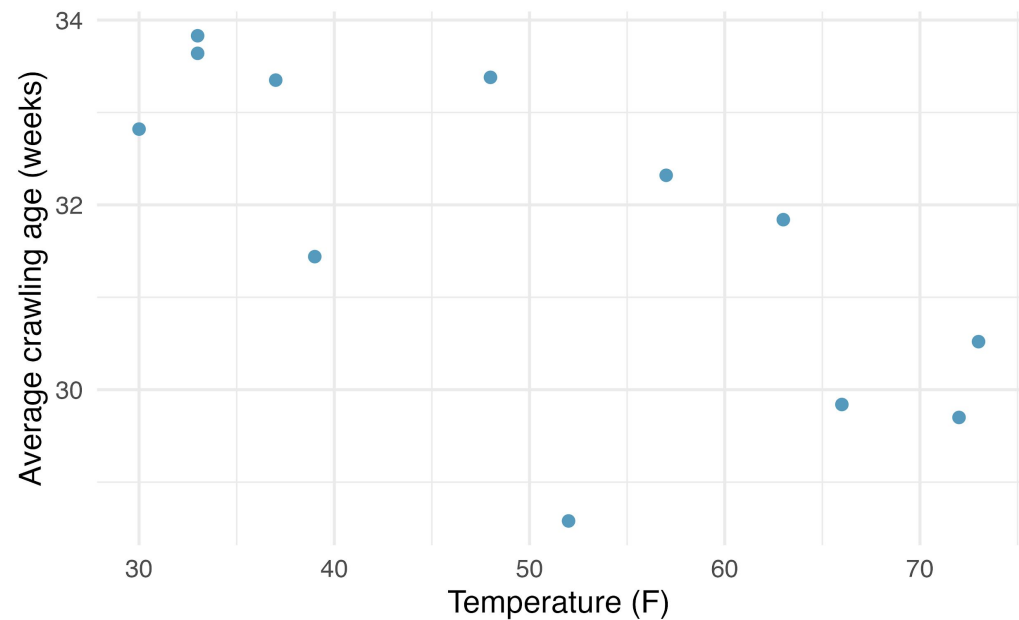
Question 2

Crawling babies. A study conducted at the University of Denver investigated whether babies take longer to learn to crawl in cold months, when they are often bundled in clothes that restrict their movement, than in warmer months. Infants born during the study year were split into twelve groups, one for each birth month. We consider the average crawling age of babies in each group against the average temperature when the babies are six months old (that's when babies often begin trying to crawl). Temperature is measured in degrees Fahrenheit (F) and age is measured in weeks. (Benson 1993)



Describe the relationship between average crawling age and average temperature as displayed in the scatterplot. Make sure to refer to the direction, strength and shape of the association, if any, and to speak in context.

The scatter plot show a relationship between the average crawling age for babies and temperature they were exposed to during infancy. Generally, there is a weak linear negative association between these two variables. The colder the temperature the higher the crawling age is for a baby, and warmer the temperature the lower the crawling age is for a baby. For example, when it was 30 degrees it took a baby around 33 weeks to begin crawling and when it was 65 degrees it took a baby around 30 weeks to begin crawling. The relationship is weak, which means there is little difference between crawling age and temperature.



More space here, if needed!